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BBACCapDefTopic

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About the BBAC Capture

About the BBAC Capture

The Broadband Analog Channel Capture (BBACCap) instrument is designed to capture waveforms for applications such as audio converters, cellular baseband, and xDSL AFEs (analog front-ends). Typical tests that can be performed include: gain versus frequency, distortion, signal-to-noise ratio, and multitone power ratio.

The following topics are included:

- | | |
|---|--|
| • | Features |
| • | Safety Information |
| • | Key Terms |
| • | BBAC Capture Software Licensing Requirements |

BBAC Capture information includes the following sections:

- | | |
|---|--|
| • | BBAC Capture Theory of Operation |
| • | BBAC Capture Programming |
| • | BBAC Capture Debug Display |
| • | BBAC Capture Examples |

Related Topics

- | | |
|---|--|
| • | BBAC15 Specifications |
| • | BBACCapture (VBT syntax) |
| • | BBAC DIB Design Guidelines |

Two BBAC Capture instruments share the BBAC instrument board with a pair of BBAC Source instruments, which generate AC waveforms. For more information, see [About the BBAC Source](#).

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About the BBAC Capture : Features

Features

The BBAC Capture instrument can accurately acquire differential signals with frequency components from DC to 15MHz. While the BBAC Capture internal processing is performed at 100Msps, the instrument’s DSP capabilities digitally resample the captured data to the user-specified sample rate in real time. The range of sample rates you can program is 5ksps to 50Msps. The BBAC Capture also has programmable voltage ranges and DC offset functionality for canceling out common mode or DC bias voltages.

The BBAC Capture offers:

- Selectable input ranges
- Programmable sample rates of 5ksps to 50Msps
- Programmable DC offset removal
- Mixed-signal pattern microcode control
- Access to Pin Parametric Measurement Unit (PPMU) per pin
- Parameter sets (PSets) for pattern control of instrument settings
- DIB Access connectivity

For BBAC Capture hardware specifications, see BBAC15 Specifications.

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About the BBAC Capture : Safety Information

Safety Information

When operating a BBAC Capture instrument, always follow general UltraFLEX test system safety guidelines. See [Test System Safety](#).

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About the BBAC Capture : Key Terms

Key Terms

AFE	Analog front-end
BBAC	Broadband Analog Channel
CCC	Channel Card Common
CMEM	Capture Memory
DGS	Device Ground Sense
HSD	High Speed Digital
IIM	Instrument Initiated Move
ISL	Instrument Sync-Link
ksps	Kilosamples per second
Msp	Megasamples per second
PPMU	Pin Parametric Measurement Unit
PSet	Parameter Set

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About the BBAC Capture : BBAC Capture Software Licensing Requirements

BBAC Capture Software Licensing Requirements

BBAC instrument hardware can support multiple frequencies across source and capture instruments. However, IG-XL enforces a software licensing scheme that constrains instrument functionality to a specified frequency limit across source instruments and a specified frequency limit across all capture instruments.

When purchasing a software license, you must specify a source bandwidth of 3MHz or 15MHz, a capture bandwidth of 3MHz or 15MHz, and the number of channels in increments of two beyond the first channel (0, 1, 2, 4, 6, 8... 48). The bandwidth value specified applies to all channels of the same type.

For example:

- | |
|--|
| <ul style="list-style-type: none">Source bandwidth: 15MHz
of channels: 8 |
| <ul style="list-style-type: none">Capture bandwidth: 3MHz
of channels: 4 |

 Note:

The maximum number of source or capture channels available (each) is 2 x (# of available test head slots), or 48 - (2 x # of occupied test head slots).

IG-XL checks for a license on validation of a test program to verify that the number of channels used in the test does not exceed the number licensed. A validation error occurs if more instruments are used in a test than are specified in the license.

IG-XL also verifies the license on execution of a test to verify that the test is not programmed to source or capture signals whose frequency exceeds the licensed limitation. A run-time error occurs

if a test sources or captures at frequencies higher than specified by the license.

A BBAC Source 3MHz license limits the source sample rate to a maximum value of 60Msps. This limits the maximum frequency that can be sourced to 3MHz, due to the requirement that the source sample rate must be 20 times greater than the highest frequency of interest. See [BBAC15 Specifications](#) for the minimum sample rate specifications.

A BBAC Capture 3MHz license limits the capture sample rate to a maximum value of 8Msps. This limits the maximum frequency that can be captured to 3MHz, due to the requirement that the capture sample rate must be 8/3 times greater than the highest frequency of interest. See [BBAC15 Specifications](#) for the minimum sample rate specifications.

License limitations are not enforced when a checker program or calibration routine is running or when a program is running on a simulator.

✓ Note:

BBAC software licenses are node-locked licenses, not floating licenses, that apply to instruments in a specific test system.

Summary of Software Licensing

- All source channels must be the same bandwidth option.
- All capture channels must be the same bandwidth option.
- Source bandwidth is independent of capture bandwidth.
- The number of source channels is limited to increments of two beyond the first instrument (0, 1, 2, 4, 6, 8... 48).
- The number of capture channels is limited to increments of two beyond the first instrument (0, 1, 2, 4, 6, 8... 48).
- Source and capture enabling is not limited. You can order all capture channels and no sources or all source channels and no captures.

For more information, see [IG-XL Licensing](#) in the IG-XL Administration section.



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BBAC Capture Examples

BBAC Capture Examples

This section contains program examples for the BBAC Capture instrument. These IG-XL workbook examples show all IG-XL data and VBT code necessary for a complete running program.

The following topics are available:

- [BBAC Source/Capture Loopback](#)
- [BBAC Capture Signals VBT Examples](#)
- [BBAC Capture PSets VBT Examples](#)

For more information, see:

- [BBACCapture](#) (VBT Syntax Reference)
- [BBAC Capture Programming](#)

	
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BBAC Capture Examples : BBAC Source/Capture Loopback

BBAC Source/Capture Loopback

This workbook example includes the following program components:

- | | |
|---|-----------------|
| • | DataTool Sheets |
| • | VBT Code |

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BBAC Capture Examples : [BBAC Source/Capture Loopback](#) : DataTool Sheets

DataTool Sheets

This workbook example uses the following DataTool sheets:

- [Pin Map Sheet](#)
- [Channel Map Sheet](#)
- [Test Instances Sheet](#)
- [Flow Table Sheet](#)
- [Time Sets \(Basic\)](#)

Pin Map Sheet

Pin Map

Group Name	Pin Name	Type
	Src1_P	Analog
	Src1_N	Analog
	Cap1_P	Analog
	Cap1_N	Analog
	DigControl	I/O

Channel Map Sheet

Channel Map

DIB ID:

View Mode:

Device Under Test		Tester Channel		
Pin Name	Package Pin	Type	Site 0	Comment
Src1_P		BBACSrcPos	4.srcpos1	4
Src1_N		BBACSrcNeg	4.srcneg1	
Cap1_P		BBACCapPos	4.cappos1	
Cap1_N		BBACCapNeg	4.capneg1	
DigControl		I/O	1.ch0	

Test Instances Sheet

Test Instances



Test Name	Test Procedure			DC Specs		AC Specs		Time Sets
	Type	Name	Called As	Category	Selector	Category	Selector	
Source_Capture_LoopBack1	VBT	Source_Capture_LoopBack						TSB

Flow Table Sheet

Flow Table



Flow Domain:

Label	Enable	Gate			Command		TName	TNum
		Job	Part	Env	Opcode	Parameter		
					Test	Source_Capture_LoopBack1		

Time Sets (Basic)

Time Sets (Basic)



Timing Mode:

Master Timeset Name:

Time Domain:

Strobe Ref Setup Name:

Time Set	Cycle	Pin/Group			Data		Drive				Compare				Edge Resolution
	Period	Name	Clock Period	Setup	Src	Fmt	On	Data	Return	Off	Mode	Open	Close	Ref Offset	Mode
ts0	200.E-09	DigControl		i/o	PAT	NR		0			Off				Auto

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[BBAC Capture Examples](#) : [BBAC Source/Capture Loopback](#) : VBT Code

VBT Code

Public Function Source_Capture_LoopBack()

```
Dim SourceSine As New DSPWave
```

```
Dim CaptureWave As New DSPWave
```

```
Dim i As Long
```

```
' Connect BBAC Source to Capture Loopback
```

```
TheHdw.BBACSource.Pins("Src1_P").Connect tIBBACSourceFromPOS Or _  
tIBBACSourceFromNEG, tIBBACSourceToLoopBack
```

```
TheHdw.BBACSource.Pins("Src1_P").Connect tIBBACSourceFromREF, _  
tIBBACSourceToDGS
```

```
TheHdw.BBACCapture.Pins\("Cap1\_P"\).Connect tIBBACCaptureFromPOS Or \_  
tIBBACCaptureFromNEG, tIBBACCaptureToLoopBack
```

```
' Create Sinewave Samples to Source
```

```
Call SourceSine.CreateConstant(0, 65536)
```

```
For i = 0 To 65536 - 1
```

```
SourceSine.Element(i) = Sin((3 * i * 2 * Pi / 65536)) / 2
```

```
Next i
```

```
' Load Setting and Samples into the BBAC and Set up Source Signal
```

```
TheHdw.BBACSource.Pins("Src1_P").Signals.Add "SourceSignal"
```

```
TheHdw.BBACSource.Pins("Src1_P").Signals.DefaultSignal = "SourceSignal"
```

```
TheExec.WaveDefinitions.CreateWaveDefinition "WaveDef", SourceSine, True
```

```
With TheHdw.BBACSource.Pins("Src1_P").Signals("SourceSignal")
```

```
.WaveDefinitionName = "WaveDef"
```

```
.Amplitude = 1
```

```
.Offset = 0
```

```
.SampleSize = 65536
.SampleRate = 655360
.CycleCount = 1
.LoadSettings
End With
' Set up BBAC Capture
TheHdw.BBACCapture.Pins("Cap1_P").Signals.Add "CaptureSignal"
TheHdw.BBACCapture.Pins("Cap1_P").Signals.DefaultSignal = "CaptureSignal"
With TheHdw.BBACCapture.Pins("Cap1_P").Signals("CaptureSignal")
    .SampleSize = 65536
    .SampleRate = 655360
    .CycleCount = 1
    .LoadSettings
End With
' Start Sourcing the signal and capturing with Pattern
'TheHdw.Digital.ApplyLevelsTiming True, True, True
TheHdw.Patterns("./src_cap_loop.pat").Load
TheHdw.Digital.Patgen.TimeoutEnable = False
TheHdw.Patterns("./src_cap_loop.pat").Start "", ""
TheHdw.Digital.Patgen.HaltWait
' Wait for Capture to complete
While TheHdw.BBACCapture.Pins("Cap1_P").IsCaptureInProgress
Wend
' Bind capture data to a DSPwave object
CaptureWave = TheHdw.BBACCapture.Pins("Cap1_P").Signals("CaptureSignal").DSPWave
CaptureWave.Plot "BBAC LOOPBACK"
' Do an SNR Analysis on captured data and display it in immediate window
Debug.Print CaptureWave.Spectrum.CalcSNR
' Disconnect the BBAC Source from the Capture
TheHdw.BBACSource.Pins("Src1_P").Disconnect tlBBACSourceFromPOS Or _
    tlBBACSourceFromNEG, tlBBACSourceToLoopBack
TheHdw.BBACSource.Pins("Src1_P").Disconnect tlBBACSourceFromREF, _
    tlBBACSourceToDGS
TheHdw.BBACCapture.Pins("Cap1_P").Disconnect tlBBACCaptureFromPOS Or _
    tlBBACCaptureFromNEG, tlBBACCaptureToLoopBack
End Function
```

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BBAC Capture Examples : BBAC Capture Signals VBT Examples

BBAC Capture Signals VBT Examples

The following examples demonstrate BBAC Capture VBT methods and properties:

- | | |
|--|---|
| | • Use MST to Set Up, Load, and Trigger a BBAC Capture Signal |
| | • Report Information About a Collection of BBAC Capture Signals |
| | • Set All BBAC Capture Hardware Signal Parameters to Use Loaded Value |
| | • Use Direct Programming to Set Up and Trigger a BBAC Capture Signal |
| | • Use LoadSettings for a BBAC Capture Signal and Read the Programmed Settings |
| | • Use Pattern Control with BBAC Capture Default Signal |

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[BBAC Capture Examples](#) : [BBAC Capture Signals VBT Examples](#) : Use MST to Set Up, Load, and Trigger a BBAC Capture Signal

[Use MST to Set Up, Load, and Trigger a BBAC Capture Signal](#)

This code uses a [Mixed Signal Timing \(MST\)](#) set to set up, load, and trigger a BBAC Capture signal.

```
Public Sub UseMSTToSetupAndTriggerBBACCapSignal()  
    ' Use MST to set up, load, and trigger a capture.  
    ' Replace "AOUT" with your bbac capture (pos) pin name  
    Dim pinName As String  
    pinName = "AOUT"  
    ' This example uses a PinListData to bind the capture DSPWave  
    Dim myDspWave As New DSPWave  
    TheHdw.BBACCapture.pins(pinName).Signals.Add "cap1"  
    With TheHdw.BBACCapture.pins(pinName).Signals("cap1")  
        ' Apply a Mixed Signal Timing Set to this Test Instance before  
        ' ApplyMixedSignalTiming is called.  
        .ApplyMixedSignalTiming  
        ' Assign the PinListData object for access to the captured data  
        myDspWave = .DSPWave  
        ' Use defaults for all other Signal properties  
        ' Load Signal hardware values into hardware  
        .LoadSettings  
        ' Trigger the Signal capture  
        .Trigger  
    End With  
End Sub
```

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[BBAC Capture Examples](#) : [BBAC Capture Signals VBT Examples](#) : Report Information About a Collection of BBAC Capture Signals

Report Information About a Collection of BBAC Capture Signals

This code reports information about a collection of BBAC Capture signals using Count and List.

```
Public Sub ReportInfoAboutBBACCapSignalsColl()  
    ' Replace "AOUT" below with your bbac capture (pos) pin name  
    Dim pinName As String  
    pinName = "AOUT"  
    ' First, add 10 Signals to the Signals collection on Pin 'pinName'  
    Dim i As Long  
    Dim sigCount As Long  
    For i = 1 To 10  
        TheHdw.BBACCapture.pins(pinName).Signals.Add "signal" & CStr(i)  
    Next i  
    sigCount = TheHdw.BBACCapture.pins(pinName).Signals.Count  
    ' Print out each Signal name  
    Dim sigList() As String  
    sigList = TheHdw.BBACCapture.pins(pinName).Signals.List  
    Debug.Print "There are " & sigCount & " signals in the collection " & _  
        "on pin '" & pinName & "'. They are:"  
    For i = 0 To sigCount - 1  
        Debug.Print vbCrLf & sigList(i)  
    Next i  
End Sub
```

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[BBAC Capture Examples](#) : [BBAC Capture Signals VBT Examples](#) : Set All BBAC Capture Hardware Signal Parameters to Use Loaded Value

[Set All BBAC Capture Hardware Signal Parameters to Use Loaded Value](#)

[This code explicitly sets all hardware signal parameters to use the loaded value.](#)

```
Public Sub SetAllBBACCaptureSignalHardwareToUseLoaded()
```

```
    ' Instruct all instrument parameters to "Use Loaded Value"
```

```
    ' Replace "AOUT" with your bbac capture (pos) pin name
```

```
    Dim pinName As String
```

```
    pinName = "AOUT"
```

```
    TheHdw.BBACCapture.pins(pinName).Signals.Add "cap1"
```

```
    With TheHdw.BBACCapture.pins(pinName).Signals("cap1")
```

```
        .BlockCount.Mode = tlSignalModeUseLoadedValue
```

```
        .BlockSpacing.Mode = tlSignalModeUseLoadedValue
```

```
        .Coupling.Mode = tlSignalModeUseLoadedValue
```

```
        .ExtraSamples.Mode = tlSignalModeUseLoadedValue
```

```
        .Offset.Mode = tlSignalModeUseLoadedValue
```

```
        .SampleRate.Mode = tlSignalModeUseLoadedValue
```

```
        .SampleSize.Mode = tlSignalModeUseLoadedValue
```

```
        .VoltageRange.Mode = tlSignalModeUseLoadedValue
```

```
    End With
```

```
End Sub
```

	
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[BBAC Capture Examples](#) : [BBAC Capture Signals VBT Examples](#) : Use Direct Programming to Set Up and Trigger a BBAC Capture Signal

[Use Direct Programming to Set Up and Trigger a BBAC Capture Signal](#)

This code uses direct programming (as opposed to using loaded values) to set up and trigger a signal.

```
Public Sub CreateAndTriggerBBACCaptureSignalWithDirectProgramming()
```

```
    ' Use direct timing programming to set up, load, and trigger a capture
```

```
    ' Replace "AOUT" with your bbac capture (pos) pin name
```

```
    Dim pinName As String
```

```
    pinName = "AOUT"
```

```
    Dim myDSPWave As New DSPWave
```

```
    TheHdw.BBACCapture.pins\(pinName\).Signals.Add "cap1"
```

```
    With TheHdw.BBACCapture.pins\(pinName\).Signals\("cap1"\)
```

```
        .SampleRate = 20000
```

```
        .SampleSize = 20
```

```
        ' Assign a PinListData object for access to the captured data
```

```
        myDSPWave = .DSPWave
```

```
        ' Use defaults for all other Signal properties
```

```
        ' Load hardware settings for the Signal
```

```
        .LoadSettings
```

```
        ' Trigger the capture
```

```
        .Trigger
```

```
    End With
```

```
End Sub
```

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[BBAC Capture Examples](#) : [BBAC Capture Signals VBT Examples](#) : Use LoadSettings for a BBAC Capture Signal and Read the Programmed Settings

[Use LoadSettings for a BBAC Capture Signal and Read the Programmed Settings](#)

This code programs a new signal, uses Load Settings, and reads the programmed hardware settings.

```
Public Sub ProgramBBACCaptureSignalAndReadbackHardware()
```

```
    ' Program a new signal, Load Settings, and read the programmed hardware
```

```
    ' settings from Pins language.
```

```
    ' Replace "AOUT" with your bbac capture (pos) pin name
```

```
    Dim pinName As String
```

```
    pinName = "AOUT"
```

```
    TheHdw.BBACCapture.pins(pinName).Signals.Add "cap1"
```

```
    With TheHdw.BBACCapture.pins(pinName).Signals("cap1")
```

```
        .SampleRate = 20000
```

```
        .SampleSize = 512
```

```
        .VoltageRange = 5.3
```

```
        .Offset.Enable = tlBBACCaptureOffsetDisable
```

```
        ' Use defaults for everything else.
```

```
        ' Load hardware settings
```

```
        .LoadSettings
```

```
    End With
```

```
    ' Now that settings are loaded, read back hardware values from Pins
```

```
    ' language
```

```
    With TheHdw.BBACCapture.pins(pinName)
```

```
Debug.Print "Loaded SampleRate: " & CStr(.SampleRate)
Debug.Print "Loaded SampleSize: " & CStr(.SampleSize)
Debug.Print "Loaded Offset Enable: " & CStr(.Offset.Enable)
Debug.Print "Loaded VoltageRange: " & CStr(.VoltageRange)
End With
```

End Sub

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[BBAC Capture Examples](#) : [BBAC Capture Signals VBT Examples](#) : Use Pattern Control with BBAC Capture Default Signal

Use Pattern Control with BBAC Capture Default Signal

This code uses pattern control with the DefaultSignal.

```
Public Sub UseBBACCapDefaultSignalAndPatternControl()
    ' Replace "AOUT" with your bbac capture (pos) pin name
    Dim pinName As String
    pinName = "AOUT"
    ' Replace ".\Patterns\myPat.pat" with your pattern name
    Dim patternName As String
    patternName = ".\Patterns\myPat.pat"
    ' Add 5 Signals to the Signals collection on Pin 'pinName'
    Dim i As Long
    Dim sigCount As Long
    Dim sigList() As String
    Dim myDSPWave As New DSPWave
    For i = 1 To 5
        TheHdw.BBACCapture.pins(pinName).Signals.Add "signal" & CStr(i)
        With TheHdw.BBACCapture.pins(pinName).Signals("signal" & CStr(i))
            ' Apply a Mixed Signal Timing Set to this Test Instance before
            ' ApplyMixedSignalTiming is called.
            .ApplyMixedSignalTiming
            ' Assign a PinListData object for access to the captured data
            myDSPWave = .DSPWave
        End With
    Next i
```

```
' Set each Signal to be default signal and burst pattern.  
For i = 1 To 5  
  With TheHdw.BBACCapture.pins(pinName).Signals  
    ' Load Settings  
    .Item("signal" & CStr(i)).LoadSettings  
    ' Set the DefaultSignal  
    .DefaultSignal = "signal" & CStr(i)  
  End With  
  ' This pattern should have an unnamed "Trig" microcode on pin  
  ' 'pinName'.  
  ' Run "" will start pattern from the beginning and wait for  
  ' completion.  
  TheHdw.Digital.Patterns.Pat(patternName).Run ""  
Next i  
End Sub
```


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BBAC Capture Examples : BBAC Capture PSets VBT Examples

BBAC Capture PSets VBT Examples

The following examples show the use of BBAC Capture PSet syntax.

Using PSets with the Block Capture Model

The following example programs BBAC Capture PSet parameters using the block capture model.

```
Public Function CreateBBACCapPSeta(CapPin As String, PSetName As String, _
    BlockCount As Long, BlockSize As Long, BlockSpacing As Long, _
    ExtraSamples As Long, SampleRate As Double, VoltageRange As Double) _
    As Long
    ' Add the PSet to the BBAC Capture PSets collection
    TheHdw.BBACCapture.Pins(CapPin).PSets.Add PSetName
    ' Set all the parameters of the PSet(Block Count, Block Size,
    ' Block Spacing, Extra Samples,Sample Rate,Voltage Range)
    TheHdw.BBACCapture.Pins(CapPin).PSets(PSetName).Set BlockCount, _
        BlockSize, _
        BlockSpacing, _
        ExtraSamples, _
        SampleRate, _
        VoltageRange
```

End Function

Using PSets for a Single Capture

The following example programs BBAC Capture PSet parameters using a single capture model.

```
Public Function CreateBBACCapPSetb(CapPin As String, PSetName As String, _
    SampleSize As Long, SampleRate As Double, VoltageRange As Double) _
    As Long
```

```
' Add the PSet to the BBAC Capture PSets collection
TheHdw.BBACCapture.Pins(CapPin).PSets.Add PSetName
' Set all parameters of the PSet(Block Count, Block Size,
' Block Spacing, Extra Samples,Sample Rate,Voltage Range)
With TheHdw.BBACCapture.Pins(CapPin).PSets(PSetName)
    .BlockSize = SampleSize
    .VoltageRange = VoltageRange
    .SampleRate = SampleRate
End With
End Function
```

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BBAC Capture Debug Display

BBAC Capture Debug Display

The Broadband AC Capture (BBAC Capture) Debug Display is an interactive diagram of the BBAC Capture hardware and signal path which shows the instrument’s settings and connections in the context of a site. You can change the values, options, and settings in the display and run your test program to see the effects of the changes.

The following sections describe the display’s controls and how to use the display:

- Starting the BBAC Capture Debug Display
- BBAC Capture Debug Display Interface
- Making Channel Connections

This section describes features specific to the BBAC Capture Debug Display. For more information on using debug displays, see the following:

- Using Debug Displays in TDE
- Producing Code from Debug Displays

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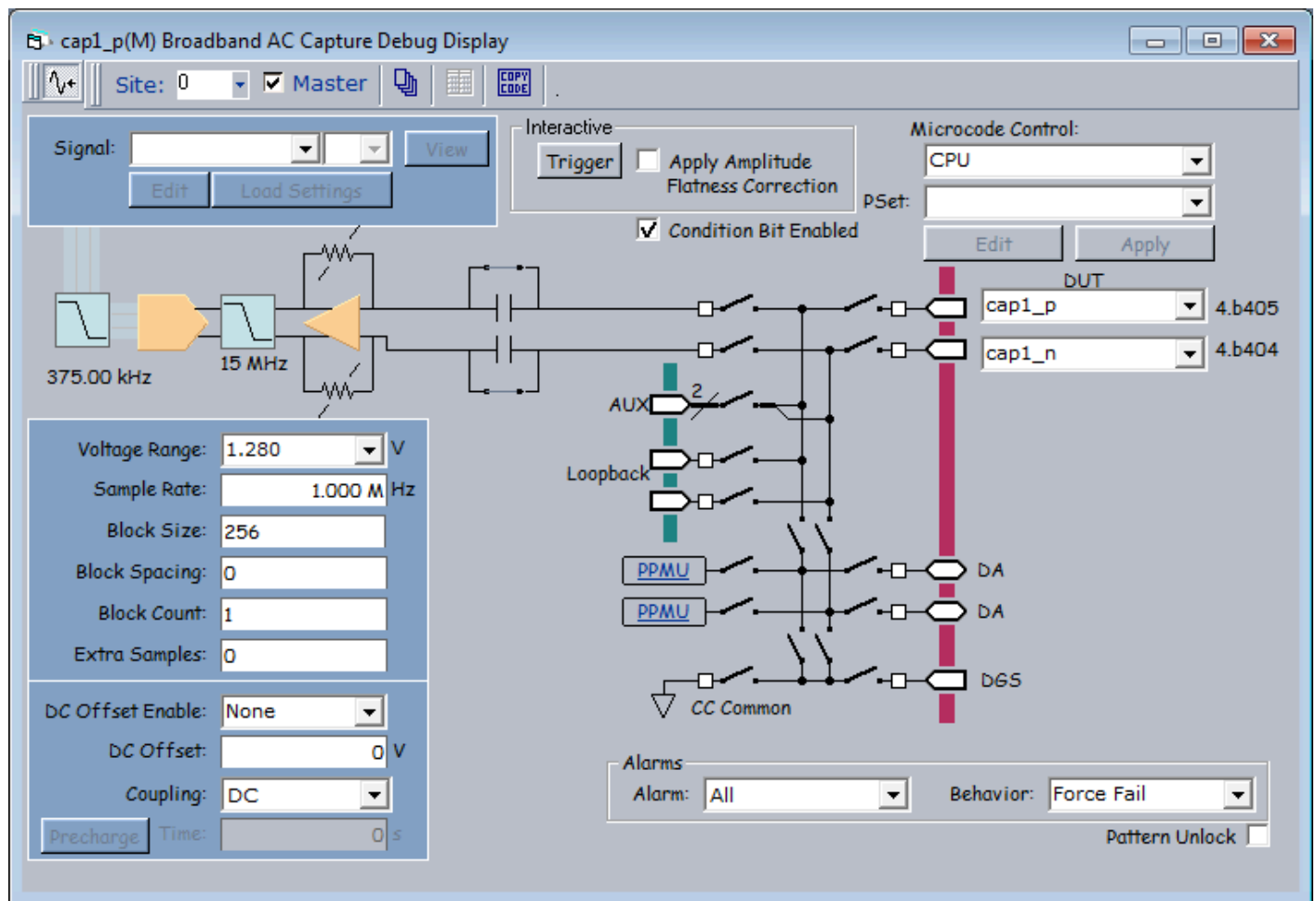


[BBAC Capture Debug Display](#) : Starting the BBAC Capture Debug Display

Starting the BBAC Capture Debug Display

To start the Broadband AC Capture Debug Display:

1. [Validate the active test program and ensure that it includes BBAC Capture pins.](#)
2. [On the IG-XL Tools tab, in the Launch group, select Debug Displays>Broadband AC Capture.](#)
This opens the display for all active sites.



Broadband AC Capture Debug Display

Incomplete Circuit or Errors Message when Starting the Debug Display

Does the Broadband AC debug display's connection matrix appear incomplete? Or, does one of the following error messages appear next to the toolbar?

! The test program has not been validated.

! No pin exists of type Broadband AC Capture.

These problems indicate that the active test program was not validated before starting the debug display, or that the program does not include BBAC Capture pins.

To correct these errors:

1. Return to the IG-XL DataTool workbook and verify that BBAC Capture pins are included and assigned in the pin map and channel map.
2. Click the Validate Job button .

If the test program successfully validates without errors, you can return to the Broadband AC Capture debug display, which will now be complete and ready for use. If the test program does not validate successfully, you must correct all errors before the debug display can be complete.

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BBAC Capture Debug Display : BBAC Capture Debug Display Interface

BBAC Capture Debug Display Interface

A trigger on the debug display functions as a trigger on an oscilloscope, in that it makes a capture using the current instrument debug display settings. Captured data from triggering the debug display are for visual reference only. You cannot use the data from the debug display capture for processing because the captured samples are not moved through IIM to the Support Board. Also, the debug display does not store the data. Once you have displayed it, therefore, you cannot retrieve it.

The table lists BBAC Capture Debug Display parameters, buttons, and fields.

BBAC Capture Debug Display Fields

Parameter/Button	Description
Signal	All user-defined signal names appear in the Signal field. When a signal is selected, the Load Settings button is enabled. When a capture is moved to the Support Board and it has not been redirected to a DSP Procedure, that signal's name is marked with an asterisk (*). You can view such a signal in WaveScope. If multiple captures on the Support Board are associated with the same Signal name, the box between the Signal field and the View button is enabled, and you can select which capture you are interested in viewing.
View	This button is enabled when you select a signal containing an asterisk. Viewing a signal only makes a copy of the samples on the Support Board. The data is not removed from the Support Board.

Edit	Brings up the PSet and Signals Editor (PSSE) with the selected signal loaded. This control is grayed out if no signals are available. See PSet and Signals Editor in the TDE section.
Load Settings	This button is enabled when a signal name is selected from the Signal field. It loads the settings of the specified Signal. That signal must be defined from VBT.
Trigger	This button functions similarly to a single trigger button on an oscilloscope. You can set the sample rate, block count, block size, block spacing, and extra samples on the display and then click Trigger to make a capture. The trigger is disassociated from the Signal name. No signal must exist for the Trigger button to function. Captures made from the display are shown on WaveScope, and the data is then discarded. The display does not move data to the Support Boards.
Apply Amplitude Flatness Correction	Allows you to improve the BBAC Capture flatness beyond the instrument's performance specification by applying DSP preprocessing can be applied to the capture signal to compensate for the capture instrument's frequency response. See Amplitude Flatness Correction .
Microcode Control	Specifies where the trigger command, which starts the capturing of the waveform, originates. The drop-down list options are: • CPU • External Trigger
PSet	Lists the parameter sets (PSets) that where defined in language. To apply a PSet from the debug display, you must have previously defined PSets using VBT. Otherwise, the drop-down list is empty.
Edit	Brings up the PSet and Signals Editor (PSSE) with the selected PSet loaded. This control is grayed out if no PSets are available. See PSet and Signals Editor in the TDE section.
Apply	Allows you to apply a PSet from the PSet list. See BBAC Capture Pattern Microcodes .

Condition Bit Enabled	Capture instrument generates a condition bit when the capture is complete. See Condition Bit Programming.
Voltage Range	The voltage range of the capture. • Minimum: 0.0V • Maximum: 10.240V
Sample Rate	The number of samples to capture per second. • Minimum: 5.000kHz • Maximum: 50.000MHz
Block Size	The number of samples per block. • Minimum: 1 • Maximum: 1 048 565 See Capture Sample Size Parameters.
Block Spacing	The number of samples to skip per block. • Minimum: 0 • Maximum: 4 294 967 295 See Capture Sample Size Parameters.
Block Count	The number of blocks to capture. • Minimum: 1 • Maximum: 4095 See Capture Sample Size Parameters.

Extra Samples

The number of extra samples per block.

- Minimum: 0
- Maximum: 1 048 304

See Capture Sample Size Parameters.

DC Offset Enable

Assigns a DC offset to the Capture pins. The options are:

- Both
- Neg
- None
- Pos

The DC offset circuitry is added to the display according to the assignment selected.

DC Offset

Sets the amount DC offset to apply.

- Minimum: -6.0V
- Maximum: 6.0V

Coupling

Selects either AC or DC coupling mode.

Precharge Time

You can only precharge in AC coupling mode. See Precharge. The precharge and Precharge button are disabled in DC coupling mode.

PPMU

Click this button to open the PPMU debug display for the selected pin. See PPMU Debug Display.

Alarm

Selects the alarm for which to set the Behavior. The options are:

- All

- [Over Range](#)

Selecting an alarm only affects the alarm for that pin. Note that All and Over Range perform the same function. Also, the special option All is write-only, so the behavior displayed for it is not updated if it is changed using VBT.

Behavior

Selects the action that should occur if the selected alarm condition exists for the pin. The options are:

- [Continue](#)
- [Default](#)
- [Force Bin](#)
- [Force Fail](#)
- [Off](#)
- [Raise Error](#)

Pattern Unlock

Allows you to use select and monitor pattern pins while that pattern is running (that is, in a CPU loop). See [PatternInstrumentUnlock](#) in the VBT Syntax Reference.

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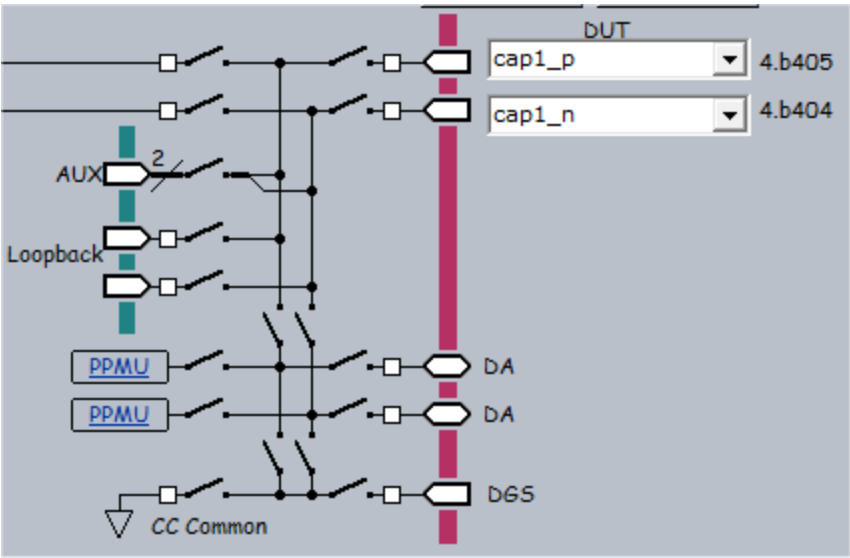
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BBAC Capture Debug Display : Making Channel Connections

Making Channel Connections

The figure shows the connections matrix on the Broadband AC Capture Debug Display. The connections matrix shows all possible BBAC Capture channel connections to available resources.



BBAC Capture Debug Display Connections Matrix

The following topics describe the interface behavior and the actions that you perform to make channel and resource connections in the Broadband AC Debug Display:

- Opening and Closing a Relay Path
- Highlighting a Closed Relay Path
- Determining a Local or Indirect Channel Connection

The following sections describe the resources that the BBAC Capture channels can be connected to:

- [Connecting DUT Pins](#)
- [Connecting DIB Access Channels](#)
- [Connecting to Device Ground Sense](#)
- [Connecting the Auxiliary Pin](#)
- [Connecting the Internal Analog Loopback Path](#)
- [Connecting a PPMU](#)
- [Connecting Channel Card Common](#)

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BBAC Capture Debug Display : Making Channel Connections : Opening and Closing a Relay Path

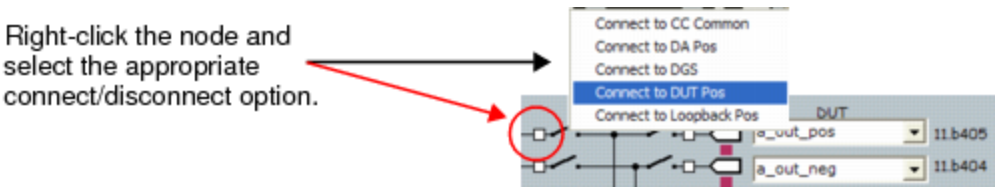
Opening and Closing a Relay Path

The Broadband AC Capture Debug Display simulates channel connections to the DUT and other resources. Graphically these connections are represented by open and closed relays along the connection path. However, clicking a relay does not open or close it. To open or close a relay path:

1.

Right-click the node along the signal path. A menu appears and lists the connect or disconnect options for that path.
2.

Select the connection destination from the menu. The relay opens or closes to make or break the connection.



Connect/Disconnect Channels and Resources Menu

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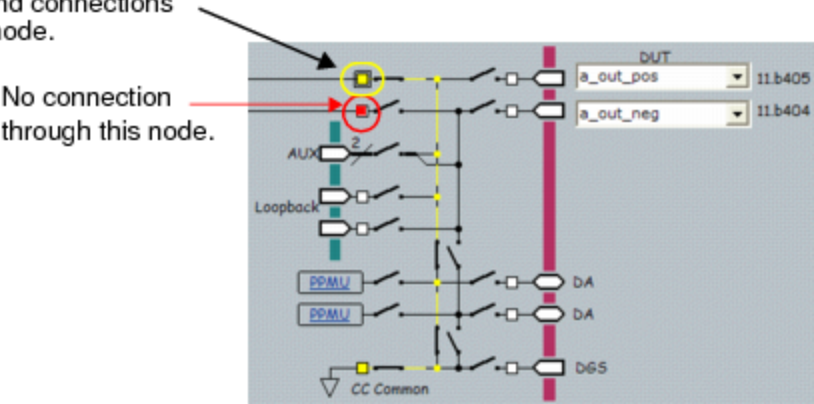
BBAC Capture Debug Display : Making Channel Connections : Highlighting a Closed Relay Path

Highlighting a Closed Relay Path

To help you determine which resources are connected to a specific BBAC Capture channel, you can highlight a connection’s relay path.

The figure shows that you can highlight a closed relay path by left-clicking and holding the node along the closed relay path. The path of closed relays and connections that run through that node are then highlighted in yellow.

Highlights the path of closed relays and connections for this node.



Tracing the Closed Relay Path

If no connection goes through that node, the node turns red.

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BBAC Capture Debug Display : Making Channel Connections : Determining a Local or Indirect Channel Connection

Determining a Local or Indirect Channel Connection

The debug display's connection matrix shows the connection paths related to BBAC Capture channels. These channel connections can be the result of connections made on this display or those made on another instrument's debug display.

For example, a BBAC Capture's DUT path may be connected to a PPMU. Although this type of connection can only be made from the PPMU display, it is reflected in both the BBAC Capture and PPMU debug displays.

When you click the DUT port of such a channel, a menu appears and lists all the pins behind that channel. The symbol next to the pin name indicates how the connection was made. A check mark next to the pin name indicates that the connection is closed from the BBAC Capture instrument to the DUT. An asterisk is next to the pin name indicates that the connection was made as a consequence of another connection.

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BBAC Capture Debug Display : Making Channel Connections : Connecting DUT Pins

Connecting DUT Pins

The Broadband AC Capture Debug Display allows you to connect an instrument channel and other resources to the DUT pins specified in the test program’s channel map.

Starting the display automatically reads in the pins and lists them in the DUT pin drop-down lists. The DIB channel string is also read into the display and is shown beside the appropriate instrument channel.

To connect an instrument channel or resource to a DUT pin:

- | | |
|----|---|
| 1. | Right-click the DUT node. |
| 2. | Select the destination resource or channel from the menu. |

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BBAC Capture Debug Display : Making Channel Connections : Connecting DIB Access Channels

Connecting DIB Access Channels

DIB access is a feature that allows a DUT pin to be connected to one instrument board through another instrument board. It is a method for connecting multiple resources on different boards to a single pin.

Since DIB access involves connecting between instruments, it also means coordinating connections between the two instrument’s debug displays. To distinguish one instrument from another, we refer to one as the “primary” instrument and the other as the “secondary.” The primary instrument is the one programmed to execute a test. The secondary instrument is the messenger or path that provides access to a specific pin.

In making a DIB access connection, you can only close the DIB Access connection from the primary instrument’s debug display.

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BBAC Capture Debug Display : Making Channel Connections : Connecting to Device Ground Sense

Connecting to Device Ground Sense

Device ground sense (DGS) is a signal that can be connected to the ground of the DUT. DGS is used as a lowside reference for the PPMUs.

To connect a DUT pin or instrument channel to the DGS:

- | | |
|----|--|
| 1. | Right-click the DGS node. |
| 2. | Select the destination pin or channel from the menu. |

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BBAC Capture Debug Display : Making Channel Connections : Connecting the Auxiliary Pin

Connecting the Auxiliary Pin

The auxiliary connection to the BBAC Capture allows another instrument to connect to the BBAC. The auxiliary is not a general purpose instrument interconnection path. It is designed to work with cabling and future product options.

When the auxiliary connection is closed, the BBAC Capture is considered a secondary instrument and the primary instrument is using the BBAC instrument. The auxiliary connections on the debug display are read-only and show whether the auxiliary connections are open or closed.

To open or close the auxiliary connection, make the connection from the primary instrument. Once the auxiliary connection is closed, no other connections on the BBAC Capture can be made.

When the auxiliary connection is closed, the Cap Pos pin is connected to AUX Pos and Cap Neg pin is connected to AUX Neg.

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BBAC Capture Debug Display : Making Channel Connections : Connecting the Internal Analog Loopback Path

Connecting the Internal Analog Loopback Path

The loopback connection is provided for debugging. It allows you to connect a BBAC Source directly to a BBAC Capture, allowing them to measure the sourced waveform.

The BBAC instrument board has two instrument pairs (BBAC Capture and BBAC Source in Channel 1 and BBAC Capture and BBAC Source in Channel 2). The BBAC Capture Positive in Channel 1 loops back to the BBAC Source Positive of Channel 1. You cannot loop back instruments between the two channels (looping BBAC Capture Positive in Channel 1 with BBAC Source Positive of Channel 2). You can control the Pos and Neg channels independently. For example, you can choose to loop back either Positive, Negative or Both.

To connect the BBAC Source to the BBAC Capture for loopback purposes:

1. Right-click the Loopback Positive node and select Connect to Cap Pos.
2. Right-click the Loopback Negative node and select Connect to Cap Neg.
3. Open the Broadband AC Source display.
4. Right-click the Loopback Positive node and select Connect to Src Pos.
5. Right-click the Loopback Negative node and select Connect to Src Neg.
6. Right-click the Source Reference node and select Connect to CCC.

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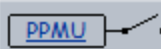
BBAC Capture Debug Display : Making Channel Connections : Connecting a PPMU

Connecting a PPMU

You can connect each BBAC Capture channel to a pin parametric measurement unit (PPMU) circuit for performing continuity tests.

Although the Broadband AC Capture Debug Display shows the channel connections to the PPMU circuits, you make PPMU connections in the PPMU debug display, not in the BBAC Capture display. To connect a BBAC Capture channel (Pos, Neg) to a PPMU circuit:

1.

 Click the appropriate PPMU hyperlink on the Broadband AC Capture Debug Display based on the BBAC Capture channel that is to be connected. The PPMU debug display starts.
2.

Select the pin to be connected to the PPMU from the Pin drop-down list.
3.

Click the relay to close the connection.

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BBAC Capture Debug Display : Making Channel Connections : Connecting Channel Card Common

Connecting Channel Card Common

You can connect a DUT pin to the channel card common (CCC) ground through either the positive or negative channels.

To connect a DUT pin or instrument channel to the CCC ground:

1. Right-click the CC Common node.
2. Select the destination pin or channel from the menu.

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BBAC Capture Programming

BBAC Capture Programming

This section describes programming of a BBAC Capture instrument to capture data from a DUT. The following topics are available:

- [Simulated Configuration Files](#)
- [Programming Concepts](#)
- [Programming Overview](#)
- [BBAC DataTool Programming](#)
- [Creating a Timing Set Using the Mixed Signal Workshop](#)
- [BBAC Capture Pattern Control](#)
- [Using BBAC in Concurrent Testing](#)

For more information, see:

- [BBACCapture](#) (VBT Syntax Reference)
- [BBAC Capture Examples](#)

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BBAC Capture Programming : Simulated Configuration Files

Simulated Configuration Files

You can use the BBAC Capture offline by forcing IG-XL to map the BBAC. Do this by creating a SimulatedConfig.txt file and placing the file in any of the following locations:

- Folder containing the IG-XL workbook
- \tester folder: C:\Program Files (x86)Teradyne\IG-XL\<version>\tester
- \bin folder: C:\Program Files (x86)\Teradyne\IG-XL\<version>\bin

The following SimulatedConfig.txt file maps the BBAC board in slot 15:

<Version>

1

</Version>

<TEST_HEAD TH 1>

#slot[.subslot]	Type	idprom (type serial rev company)
-----------------	------	----------------------------------

15.0	BBAC-15	979-214-00 0000000 0000-A 0000
	BBAC-15	956-375-00 0000000 0000-A 0000

24.0	SupportBoard	974-220-12 0000000 0000-A 0000
	SupportBoard	956-354-00 0000000 0000-A 0000

62.0	DistributionBoard	956-355-00 0000000 0000-A 0000
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To use the BBAC Capture offline, the SimulatedConfig.txt file must include every board needed in the IG-XL workbook. The SimulatedConfig.txt file represents the state of the offline tester. To use the BBAC Capture online, you need not edit any files. IG-XL automatically maps all boards in the tester.

For more information, see [Tester Configuration](#).

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BBAC Capture Programming : Programming Concepts

Programming Concepts

Conceptual information about programming methodology and test parameters is provided in the following sections:

- | | |
|---|--------------------------------|
| • | Capture Sample Size Parameters |
| • | Condition Bit Programming |
| • | Internal Analog Loopback |
| • | External Trigger |
| • | Precharge |
| • | Amplitude Flatness Correction |
| • | BBAC Capture Reset Values |

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BBAC Capture Programming : Programming Concepts : Capture Sample Size Parameters

Capture Sample Size Parameters

You can program BBAC Capture sample size for a single capture or for a block capture. When performing a single capture, use the SampleSize parameter to specify the number of samples to capture. In addition to SampleSize, capture extra samples using the ExtraSamples parameter, which can be a useful technique to manage device settling times.

The total capture size is calculated as:

Total Capture Size = SampleSize + ExtraSamples

☑ **Note:**

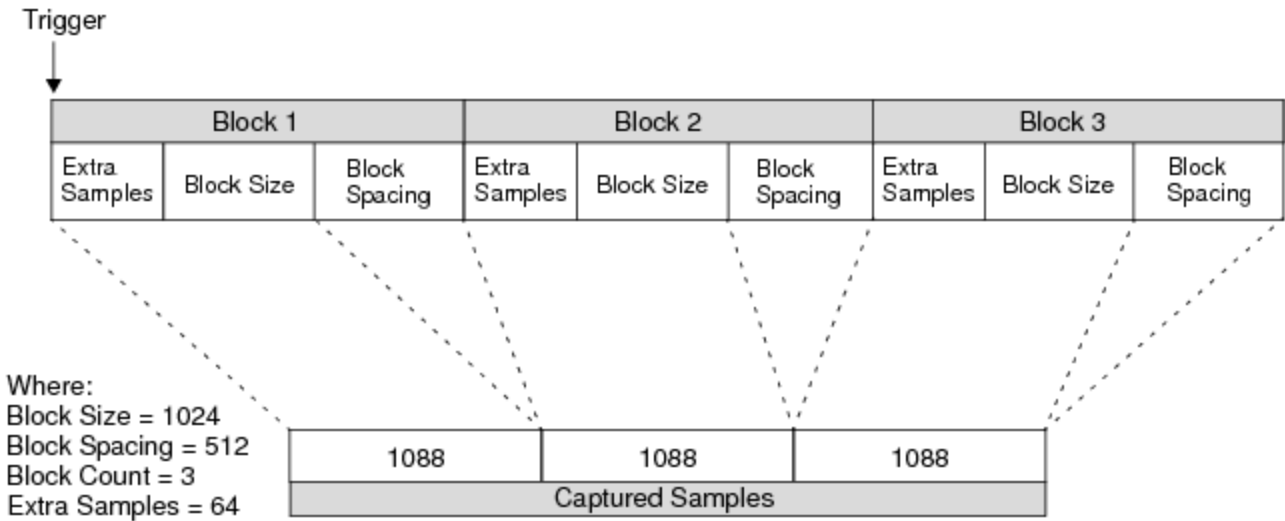
The single capture's Total Capture Size equation assumes that BlockCount equals 1, which is the default value.

When performing a block capture, the BlockSize and BlockSpacing parameters work with the BlockCount parameter to capture specified sets of samples. The BlockSize parameter specifies the number of samples in each of the one or more blocks you want to capture. The BlockSpacing parameter specifies how many samples are to be skipped (not captured in memory) between each pair of blocks captured. The BlockCount parameter specifies the number of blocks to capture. Additionally, each block can capture extra samples as specified by the ExtraSamples parameter.

The total capture size is calculated as:

Total Capture Size = (BlockSize + ExtraSamples) * BlockCount samples

This figure shows an example of using the block capture mode to capture three blocks of data.



Capture Sample Size Parameters

☒ **Note:**

The BlockSize parameter is an alias of the SampleSize parameter. The SampleSize syntax is more commonly used when block capture mode is not required.

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BBAC Capture Programming : Programming Concepts : Condition Bit Programming

Condition Bit Programming

The BBAC Capture instrument can produce a condition bit when a capture is complete. This allows a pattern to enter a capture loop so that it can keep running (producing DUT clocks, digital I/O, and so on) until the capture completes. When the capture completes, the pattern can either halt or set a flag to indicate to the host computer that the capture is finished. You can then suspend execution of the test program until the capture completes, at which time it is safe to continue on to set up the next test.

This use of the condition bit allows for flexible capture sizes and is easier than the error-prone technique of counting pattern vectors and converting into an equivalent time duration to determine which vector will be executing when the digitizer completes its capture. Another common technique is to use a fixed loop size in the pattern, but this does not always work with different capture timing solutions. A fixed loop size can exit too early if the capture is longer than expected, or it can be inefficient and run too long if the capture is shorter than expected.

Each BBAC capture can have its own microcode column in a pattern. Using the `Enable_Inst_Cond` and `Disable_Inst_Cond` microcodes, you can open and close a window that allows the end of capture instrument condition pulse to move to the pattern generator. This allows you to dynamically select which instrument delivers a condition bit during a pattern burst. Note that if the window is not open at the time when the capture completes, the pulse is missed.

In addition to pattern microcode control of the BBAC Capture instrument condition bit, you can use the following VBT code to control enabling and disabling the bit:

' Enable the capture instrument condition bit:

```
TheHdw.BBACCapture(CapPin).ConditionBitEnabled = True
```

' Disable the capture instrument condition bit:

```
TheHdw.BBACCapture(CapPin).ConditionBitEnabled = False
```

All UltraFLEX test system instrument condition bits are OR'ed together before being sent to the pattern generator. Therefore, only one instrument should have its condition bit window enabled at a time to ensure that the specific instrument has completed its capture, or reached another state that sets its condition bit.

Related Topics

- [BBAC Capture Pattern Control](#)
- [About PatternTool](#)

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BBAC Capture Programming : Programming Concepts : Internal Analog Loopback

Internal Analog Loopback

The BBAC instrument supports internal analog loopback between source and capture pairs (Source 1 and Capture 1 or Source 2 and Capture 2). This loopback feature can be used for verifying instrument performance and setup conditions without requiring a DIB or external instrumentation. Each Source and Capture instrument has an analog loopback relay path devoted to this function. You can source and capture a signal within the same instrument board, thus reducing influences outside of the instruments themselves.

The following code demonstrates connecting the BBAC Source to Capture in loopback:

```
Public Function CloseBBACLoopbackRelay(SrcPin As String, _
    CapPin As String) As Long
    TheHdw.BBACCapture.Pins(CapPin).Connect tIBBACCaptureFromInst, _
        tIBBACCaptureToLoopBack
    TheHdw.BBACSource.Pins(SrcPin).Connect tIBBACSourceFromNEG Or _
        tIBBACSourceFromPOS, _
        tIBBACSourceToLoopBack
    TheHdw.BBACSource.Pins(SrcPin).Connect tIBBACSourceFromREF, _
        tIBBACSourceToCCC
```

End Function

The following code demonstrates disconnecting the BBAC Source from Capture in loopback:

```
Public Function OpenBBACLoopbackRelay(SrcPin As String, _
    CapPin As String) As Long
    TheHdw.BBACCapture(CapPin).Disconnect tIBBACCaptureFromInst, _
        tIBBACCaptureToLoopBack
    TheHdw.BBACSource(SrcPin).Disconnect tIBBACSourceFromNEG Or _
        tIBBACSourceFromPOS, _
        tIBBACSourceToLoopBack
```

```
TheHdw.BBACSource(SrcPin).Disconnect tIBBACSourceFromREF, _  
tIBBACSourceToCCC
```

End Function

 Note:

Make sure that the BBAC Source and Capture are disconnected from other signal paths (for example, from the DIB/DUT) to avoid external signal interference while using internal loopback.





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BBAC Capture Programming : Programming Concepts : External Trigger

External Trigger

The BBAC instrument board features an external trigger pin for each BBAC instrument, for a total of four pins per board. The external trigger pin is an input pin to the BBAC instrument that allows you to start sourcing a signal or to trigger a capture based on an external event. You can use the external trigger functionality to synchronize the BBAC with other instruments in the tester by sending a signal from the DIB to any of the four AC channels on the BBAC.

Only use the external trigger functionality when you cannot use pattern control for the instrument/DUT synchronization. When using the external trigger, the start or trigger is synchronized to an external rising edge of a pulse from a digital channel that is presented to the External Trigger input of a BBAC.

You control the BBAC's external trigger pins using VBT language statements that allow you to select microcodes (like START, TRIG) to be executed when an event occurs at the trigger input.

See the [BBAC15 Specifications](#) for specific guidelines and specifications for the minimum time between events and the minimum pulse width when using BBAC external trigger pins.

For more information, see:

- [BBAC Instrument Board](#)
- [BBAC Capture Pattern Control](#)



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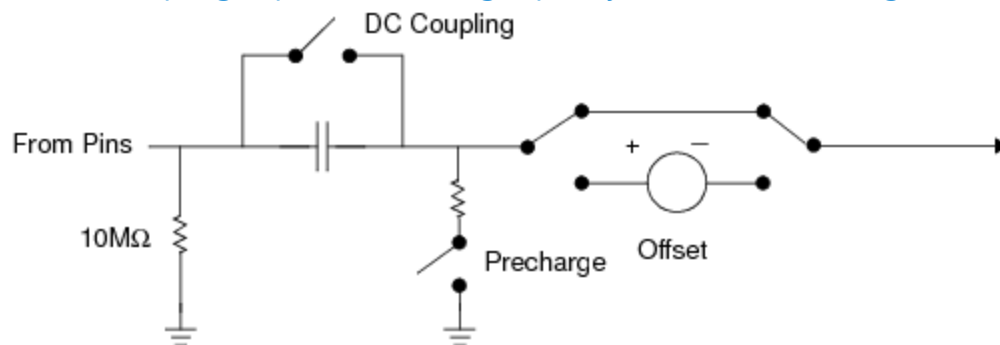
BBAC Capture Programming : Programming Concepts : Precharge

Precharge

Use the Precharge function on the BBAC Capture instrument to minimize settling time associated with using AC coupling. The BBAC Capture has a high input impedance which allows the AC coupling capacitor to be small and the low frequency corner to be low. However, this also lengthens the settling time to charge to 2ppm of final DC value. (For more information, see Coupling and Offset.)

To reduce settling time, precharge circuits are incorporated in the AC coupling paths. The precharge function works by connecting a low value resistor in parallel with the normal input resistor, allowing the coupling capacitor to rapidly charge to the DC signal component. After the capacitor has charged, the AC signal can be turned on and captured.

The figure shows that the precharge circuit provides a low impedance path to ground, which allows the AC coupling capacitor to charge quickly and reduce settling time.



Model of the Precharge Circuit with AC Coupling

For the settling time value, see the BBAC15 Specifications.

Precharge is specified as a duration (in seconds), and the precharge circuit is activated immediately and remains active for the duration programmed. It is then automatically deactivated. Note that a small transient is associated with the precharge, which must settle. For small voltage steps, the settling time improvement from the precharge is negated by the transient settling time.

✔ Note:

The same precharge circuitry is also used by the baseline removal circuitry to quickly reach the programmed DC level. When setting the voltage offset, the precharge is set automatically, but it can be disabled using VBT language. See `TheHdw.BBACCapture.Pins.Offset.AutoPreChargeEnabled`.

✔ Note:

Only use the precharge feature when the hardware is programmed for AC coupling mode.

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BBAC Capture Programming : Programming Concepts : Amplitude Flatness Correction

Amplitude Flatness Correction

To improve the BBAC Capture flatness beyond the instrument’s performance specification, DSP preprocessing can be applied to the capture signal to compensate for the capture instrument’s frequency response. The amplitude flatness correction is applied by the DSP preprocessor by performing the following steps:

1. FFT of the time domain capture data
2. Multiplication by compensation factors to apply the flatness correction
3. Inverse FFT of the frequency domain data (Optional)

Amplitude flatness correction can be applied in three ways:

No Correction	Amplitude flatness correction is not applied to the captured signal. Default.
Time Domain Correction	Amplitude flatness correction is applied to the captured signal, and the corrected time domain signal is returned.
Frequency Domain Correction	Amplitude flatness correction is applied to the captured signal, and the corrected spectrum of the signal is returned.

Note:

The frequency domain correction executes faster because an inverse FFT is not required. Because the DspWave data is in the frequency domain, however, user-DSP code may need to be modified.

[See DSP Example When Using Flatness Correction.](#)

✓ Note:

Amplitude flatness correction is only intended for coherent time domain capture signals. Since incoherent capture signals cause spectral leakage when an FFT is performed, applying the flatness correction can introduce error in the signal. Therefore, using extra samples and/or block captures may not work with capture flatness correction.

✓ Note:

Phase information is preserved when the amplitude flatness correction factor is applied.

Use the following VBT statements to program the correction mode for the capture signal:

```
TheHdw.BBACCapture.Pins(capPin).Signals.Add "signalName"
```

```
With TheHdw.BBACCapture.Pins(capPin).Signals("signalName")
```

```
    .FlatnessCorrection = Type
```

```
End With
```

Where the value of Type can be one of:

- [tIBBACCaptureFlatnessCorrectionNone](#)

- [tIBBACCaptureFlatnessCorrectionTimeDomain](#)

- [tIBBACCaptureFlatnessCorrectionFrequencyDomain](#)

DSP Example When Using Flatness Correction

```
Public Function CalcAC_Params(ByVal CapWave As DSPWave, Amplitude As Double, SNR As Double) As Long
```

```
    Dim Spectrum As New DSPWave
```

```
    If CapWave.Domain = TimeDomain Then
```

```
        ' This is the case if amplitude corrections are not enabled, or the
```

```
        ' amplitude correction type is time domain
```

```
        Spectrum = CapWave.Spectrum
```

```
    Else
```

```
        ' This is the case if the amplitude correction type is frequency domain
```

```
        Spectrum = CapWave
```

```
    End If
```

```
    Amplitude = Spectrum.CalcAmplitudeFromSpectrum
```

```
    SNR = Spectrum.CalcSNR
```

```
End Function
```

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BBAC Capture Programming : Programming Concepts : BBAC Capture Reset Values

BBAC Capture Reset Values

The table lists the local reset values for the BBAC Capture. Pre-job reset, post-job reset, power-up reset, and user reset all result in the same state.

Reset Values for BBAC Capture Settings

Setting	Reset Value
Alarms	All alarms enabled
Block Count	1
Block Size	256
Block Spacing	0
Condition Bit	Enabled
Coupling	DC
Offset	Disabled and 0.0V
External Trigger Load	The loaded command is removed.
Extra Samples	0
Microcode Control	tlBBACCaptureMicroCodeControlByCPU
Sample Rate	1MHz
Voltage Range	1.28V

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BBAC Capture Programming : Programming Overview

Programming Overview

To develop a test program that uses the BBAC Capture instrument, follow this basic procedure:

1. Create a pin map with pins of the DUT (Programming the Pin Map)
2. Create a channel map that allocates the tester resources to the DUT pins (Programming the Channel Map).
3. Create a mixed signal timing set (Creating a Timing Set Using the Mixed Signal Workshop).
4. Create a test flow, the order in which the test will execute, in the Flow Table sheet.

– Create a test function using VBT code in the Visual Basic editor.

– Instantiate the VBT test and set the test values in the Instance Editor.
5. Validate and run the test program.

For more information, see [BBACCapture](#) in the VBT Syntax Reference.

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BBAC Capture Programming : BBAC DataTool Programming

BBAC DataTool Programming

This section describes how to program several DataTool worksheets to handle basic BBAC Capture parameters. It includes the following topics:

- [Programming the Pin Map](#)
- [Programming the Channel Map](#)

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BBAC Capture Programming : [BBAC DataTool Programming](#) : Programming the Pin Map

Programming the Pin Map

A pin map describes the device pins by assigning a pin type to each device pin name. For the BBAC, the connections that you enter in the pin map are the analog channel connections to the device under test. Select the Analog pin type for each device pin name connected to a BBAC instrument. Analog is the only valid pin type for BBAC Capture pins, unless the BBAC Capture is a secondary resource. For more information, see [Pin Map Sheet](#) in the DataTool section.

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BBAC Capture Programming : BBAC DataTool Programming : Programming the Channel Map

Programming the Channel Map

A Channel Map sheet assigns individual device pin names to test system channels and power supplies. For the BBAC Capture instrument, you must select one of two channel types from the Type column menu for each pin or pin group. Valid BBAC Capture channel types are:

BBACCapNeg	BBAC Capture Negative Pin
BBACCapPos	BBAC Capture Positive Pin

Type the tester channel information value for each site to connect a DUT pin to the instrument. The table lists the available BBAC Capture channels. Note that in the table, x indicates the DIB slot number of the BBAC Capture.

BBAC Capture Channel Resources

BBAC Capture Pogo Pin	DIB Channel	Signal Name
BBAC Capture Pos 1	x.b405	x.cappos1
BBAC Capture Neg 1	x.b404	x.capneg1
BBAC Capture Pos 2	x.f405	x.cappos2
BBAC Capture Neg 2	x.f404	x.capneg2

For more information, see [Channel Map Sheet](#) in the DataTool section.

Related Topics

Information on adapting these instructions for various DIB connection options is available in the following sections:

- [Differential Device Output Connections](#)
- [Single-Ended Device Output Connections](#)
- [Shared Resources](#)
- [DIB Access](#)
- [Multiple Resources](#)

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BBAC Capture Programming : Creating a Timing Set Using the Mixed Signal Workshop

Creating a Timing Set Using the Mixed Signal Workshop

The major function of a BBAC Capture instrument is to capture analog waveforms from the DUT. The BBAC Capture instrument uses timing sets to specify capture sampling parameters such as sample rate and sample size.

The Mixed Signal Workshop (MSW) tool helps you to build reusable mixed signal timing solutions that meet coherence requirements. The output of the MSW is a mixed signal timing set which contains key capture sampling parameters such as sample rate and sample size. Timing sets are stored on a Mixed Signal Timing sheet in the IG-XL workbook and become a reusable timing solution for future tests.

To create a timing set:

1. Insert and set up a Mixed Signal Timing sheet in the IG-XL workbook.
- a. Select Insert>Insert IG-XL Worksheet in the Sheet group on the IG-XL Main tab.

b. Select Mixed Signal Timing from the Sheet type menu in the Insert IG-XL Worksheet dialog box.

c. Type a name for the sheet in the Sheet name box (optional). Default name is Mixed Signal Timing.

d. Click OK.

e. Type a timing set name in the Set Name column and press Enter.

- f. [Right-click the timing set name and select](#) Edit from the context menu to start the Mixed Signal Workshop (MSW) tool in TDE.

The MSW display appears and displays the DUT and a capture resource block.

2. [If a BBAC Capture was not automatically added to the MSW display, add a BBAC Capture resource to the MSW display. Otherwise, proceed to step 3.](#)

- a. [Select](#) MSW>Add Capture.

- b. [Select](#) BBAC Capture from the Select Resource dialog box.
Leave the Resource ID box empty, unless this timing set is using additional BBAC Capture instruments. If that is the case, assign the current capture instrument a unique ID name to distinguish it from the others. Note that the Resource ID must begin with a letter and can only contain letters, numbers, or the underscore character.

- c. [Click](#) OK in the Select Resource dialog box.
The BBAC Capture resource block is added to the MSW display.

☑ Note:

[MSW launches a blank set with a default Capture instrument resource. If you do not require this resource, select the block, then right-click. Choose](#) Delete instrument from the menu.

3. [Specify Capture timing. Typically, the objective of programming a BBAC Capture instrument is to find a sample rate \(\$F_s\$ \) and sample size \(N\) that provide the required bandwidth, frequency resolution, and \(if necessary\) signal coherence.](#)

- a. [Click the](#) Solve button in the Capture resource block to start the Solver tool.

- b. [Select the](#) F_s radio button to solve for the sample rate (F_s).

- c. [In the](#) Other Solution Criteria frame, type in the value for the minimum capture bandwidth of interest. Typically, for sine wave testing you also select the M,N Relatively Prime check box.

d. [In the](#) Number of Cycles frame, specify a range for the bin of the fundamental frequency. Typically for sine wave testing, choose the number of cycles (M) to be odd to facilitate finding a timing solution that has M and N mutually prime.

e. [In the](#) Sample Count frame, specify a range for the capture's sample size. Typically for sine wave testing, choose the number of samples (N) to be a power of two to facilitate finding a timing solution that has M and N mutually prime.

f. [In the](#) Repeat Frequency frame, specify the expected capture frequency (FR).

☑ Note:

[In many cases, the capture timing solution \(Fs and N\) actually constrains the value for a coherent source frequency \(FR\). The description in step 3 assumes that this is not the case and that the capture instrument is being programmed to coherently capture a known source frequency.](#)

g. [Click the](#) Show Choices button, select a solution from the table, and click OK. The best solution can depend on the specifics of a test's needs. Note that if no solutions are found, you must modify the inputs to the Solver. Typically, this is the sample size range or cycle count range that was used. Additionally, you can select the Show All check box to see why some of the solutions were not valid.

4. [Save the timing set by clicking the](#) Save to Excel button in the toolbar.

☑ Note:

[If you know the desired value for the capture sample rate \(Fs\) and sample size \(N\), you can program them directly into the capture resource block fields without using the Solver. When not using the Solver, be sure to specify a capture sample rate that provides sufficient bandwidth given the effect of the 3/4 band filter. For more information, see Digital Resampling and Filtering. For more information, see Mixed Signal Workshop.](#)

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BBAC Capture Programming : BBAC Capture Pattern Control

BBAC Capture Pattern Control

You can control a BBAC Capture instrument using pattern microcodes, allowing changes to instrument states and parameters during a program. You can obtain repeatable timing by programming an instrument using pattern microcodes and parameter sets (PSets). You create a pattern independently of the test program using the PatternTool or a text editor. Patterns are composed of a series of numbered digital vectors. These digital vectors define the data for each channel listed in the pattern, as well as defining the pattern microcode used for controlling the instrument during pattern execution. The following topics provide more information about BBAC Capture pattern microcodes and ASCII pattern syntax:

- [BBAC Capture Pattern Microcodes](#)
- [BBAC Capture ASCII Pattern Syntax](#)

Note the following:

- To use pattern microcodes to control the BBAC Capture on UltraFLEX, you must include a digital pin in the pin map, the channel map, and the pin list of the pattern module.
- When loading BBAC patterns, do not specify pattern module names in the VBT load command. That is, do not use something like:

TheHdw.Patterns("MyPat1.pat:module1").Load
Instead, use:

[TheHdw.Patterns\("MyPat1.pat"\).Load](#)

For more information, see:

- [Opening and Managing Pattern Files](#)

- [TheHdw.Patterns.Load](#)

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BBAC Capture Programming : BBAC Capture Pattern Control : BBAC Capture Pattern Microcodes

BBAC Capture Pattern Microcodes

A microcode controls a single function, such as starting the capture of a signal. A series of extended microcode commands are available for use inside a pattern. Patterns allow you to control the precise timing of when a signal is captured. To establish repeatable timing, or to synchronize with other instruments (such as the digital subsystem) in the tester, use a pattern. If you want to start capturing a signal but are not concerned with precise timing, however, then you do not need a pattern.

The table lists the extended microcodes used for issuing commands to a BBAC Capture instrument from a pattern.

BBAC Capture Pattern Microcodes

Microcode	Description
Trig <SignalName>	The instrument immediately starts capturing the named capture signal. This microcode takes an optional argument, which is the name of the signal to be captured. If the signal name is unspecified, the default signal is used (see TheHdw.BBACCapture.Pins.Signals.DefaultSignal).
Resync	Forces phase alignment of the BBAC capture instrument clock with the HSD T0 clock.
PSet <PSetName>	Executes the specified parameter set. The PSet microcode requires a single argument, which is the name of the PSet to be applied.

Enable_Inst_Cond	Enables the instrument condition. The BBAC Capture asserts the instrument condition to the pattern generator at the completion of a capture.
Disable_Inst_Cond	Disables the BBAC Capture from asserting the instrument condition.
Enable_Alarm	Enables all BBAC Capture alarms.
Disable_Alarm	Disables all BBAC Capture alarms.

 Note:

BBAC microcodes have specific rules regarding the necessary spacing in vectors between microcodes on a specific channel. For more information, see BBAC15 Specifications.

A parameter set (PSet) is a unique microcode that contains a set of values that describe a specific instrument setup. PSets allow a test program to change instrument setups within a pattern execution. Although a PSet is a hardware capability that changes the setup of an instrument during a pattern burst, you define it using VBT language and invoke it using PSet microcode. See BBAC Capture PSets VBT Examples.

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[BBAC Capture Programming](#) : [BBAC Capture Pattern Control](#) : [BBAC Capture ASCII Pattern Syntax](#)

BBAC Capture ASCII Pattern Syntax

BBAC Capture pattern microcodes are instructions you can place on an individual vector of a pattern. For that vector, a microcode instructs the BBAC Capture to perform a particular action, such as triggering a new capture.

Capture microcodes can be used only with pins that the instruments statement has specified as BBACCapture. For example:

```
instruments = {
    DUT_AOUT:BBACCapture;
}
```

For more information, see [Instruments Statement](#) in the Pattern Language section.

If a vector contains any BBAC Capture related microcode other than nop, that vector is placed in SRM.

The format for specifying a BBAC Capture microcode on a vector is:

(pin-list : BBACCapture = microcode)

The following code shows the ASCII pattern microcode syntax for BBAC Source and Capture instruments based on the pattern in the figure below. The device analog input (DUT_AIN) and analog output (DUT_AOUT) are connected to a BBAC Source and a BBAC Capture respectively.

```
instruments = {
    DUT_AOUT:BBACCapture;
    DUT_AIN:BBACSource;
}
vector ( $tset, Digital1, Digital2)
{
    > tset1 0      1      ;
```

```

(DUT_AOUT = Resync)
(DUT_AIN = Resync)
> - 1 1 ;
(DUT_AOUT = Trig )
(DUT_AIN = Start )
> - 0 1 ; /* Trigger/Start the default signal */
(DUT_AOUT = Trig Audio_1kHz)
(DUT_AIN = Start Audio_1kHz)
> - 1 1 ; /* Trigger/Start a named signal */
(DUT_AOUT = Enable_Alarm)
(DUT_AIN = Start1 )
> - 0 1 ;
(DUT_AOUT = Disable_Alarm)
(DUT_AIN = StartE )
> - 1 1 ;
(DUT_AOUT = Enable_Inst_Cond)
(DUT_AIN = Start1E )
> - 0 - ;
(DUT_AOUT = Disable_Inst_Cond)
(DUT_AIN = Stop)
> - 1 - ;
(DUT_AIN = StopE)
> - 0 - ;
(DUT_AOUT = PSet AudioPSet)
(DUT_AIN = PSet AudioPSet)
> - 1 - ;
> - - - ;
}

```

Comment - 4

TSet	Command	Label	Vector/Cycl	DUT_AOUT (BBACapture)	DUT_AIN (BBACSource)	Digital1	Digital2	Comment
tset1			0			0	1	
-			1	Resync	Resync	1	1	
-			2	Trig	Start	0	1	Trigger / Start the default signal
-			3	Trig Audio_1kHz	Start Audio_1kHz	1	1	Trigger / Start a named signal
-			4	Enable_Alarm	Start1	0	1	
-			5	Disable_Alarm	StartE	1	1	
-			6	Enable_Inst_Cond	Start1E	0	-	
-			7	Disable_Inst_Cond	Stop	1	-	
-			8		StopE	0	-	
-			9	PSet AudioPSet	PSet AudioPSet	1	-	
-			10			-	-	

Site : 0 (IA) No pattern started Stopped at Vec #: 0, Cycle #: 0 Stopped Not Loaded Pass Vertical: 11 Vectors Vectors 0 to 10

BBAC Pattern Microcodes Example

 Note:

The pattern microcodes in this figure are spaced too closely to be executed by the Pattern Generator. The minimum spacing required by the pattern generator depends on the BBAC sample rate and the pattern vector rate. For minimum spacing requirements, see BBAC15 Specifications.

Related Topics

- | | |
|---|--|
| • | Opening and Managing Pattern Files |
| • | Pattern Compiler |
| • | Pattern Reverse Compiler |

	
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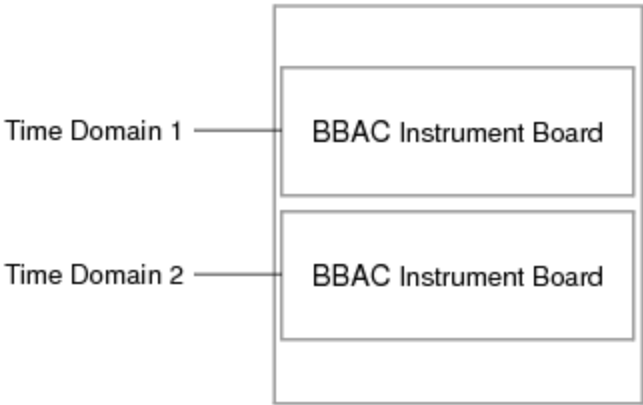
BBAC Capture Programming : Using BBAC in Concurrent Testing

Using BBAC in Concurrent Testing

To support concurrent testing, you can assign each BBAC instrument to a unique time domain. Concurrent features for BBAC include:

- Each BBAC board can be configured for one time domain
- Each BBAC instrument can be controlled by a unique time domain using Sync-Link. Features include:
 - Independent access to Sync-Link
 - Independent sample clock programming, one sample rate per channel
 - Independent clock synchronization with other BBAC instruments or with digital clocks
 - Independent control by digital pattern
 - Independent capture memory and triggers
- Multiple instruments can be grouped into a common time domain
- BBAC is restricted to using the Global flow domain and synchronous pattern starts

The following figure shows a configuration with each BBAC instrument assigned to its own time domain.



BBAC Concurrent Testing Configuration Example, Two Time Domains
For more information, see [About Concurrent Test](#).

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BBAC Capture Theory of Operation

BBAC Capture Theory of Operation

Conceptual and reference information about the BBAC Capture instrument’s hardware and signal path is provided in the following sections:

- [BBAC Instrument Board](#)
- [BBAC Capture Signal Path](#)
- [Connectivity Options](#)
- [User View of DIB Pad Patterns](#)

For information on BBAC Capture instrument performance specifications and standards, see [BBAC15 Specifications](#).

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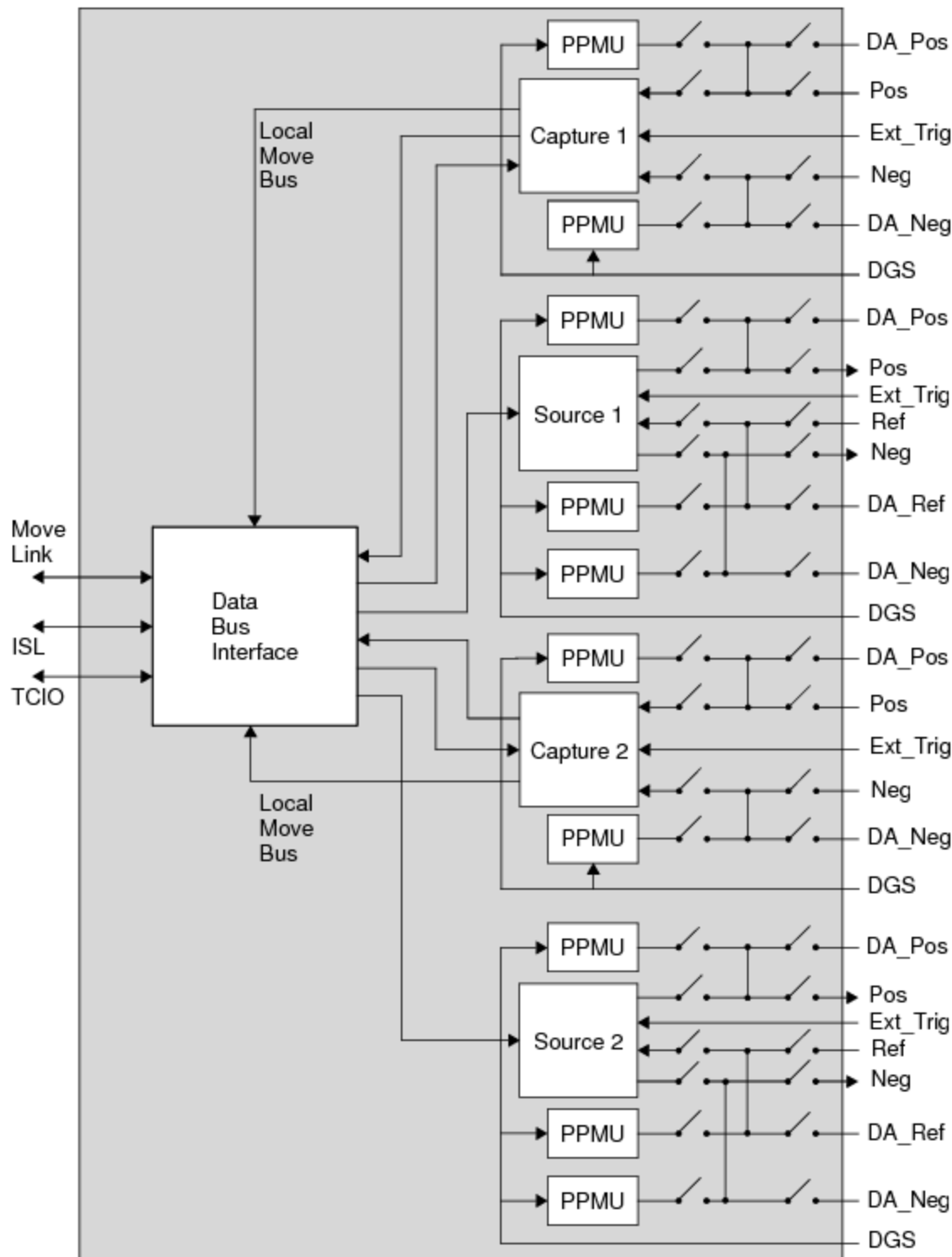
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[BBAC Capture Theory of Operation](#) : BBAC Instrument Board

BBAC Instrument Board

The figure shows a block diagram of the BBAC instrument board. The board occupies a single slot in the UltraFLEX test head and contains two BBAC Source instruments and two BBAC Capture instruments.



BBAC Instrument Board

Features of the BBAC instrument board include:

- 10 Pin Parametric Measurement Unit (PPMU) instruments, which can be used for parametric testing
- MoveLink connections from the BBAC Capture instruments to one or more dedicated DSP computers in the test system for high speed transfer and DSP processing of acquired capture

	data
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- | |
|--|
| <ul style="list-style-type: none">Instrument Sync-Link (ISL) connections from each BBAC instrument to the Pattern Generator in the High Speed Digital instrument for accurate and repeatable synchronization |
| <ul style="list-style-type: none">DIB Access connections per instrument channel |
| <ul style="list-style-type: none">An internal analog loopback connection between each BBAC Source and Capture instrument pair |
| <ul style="list-style-type: none">External trigger connections for starting a BBAC Source and triggering a BBAC Capture |
| <ul style="list-style-type: none">One Device Ground Sense (DGS) connection shared by the PPMU circuits for each Source and Capture channel |

	
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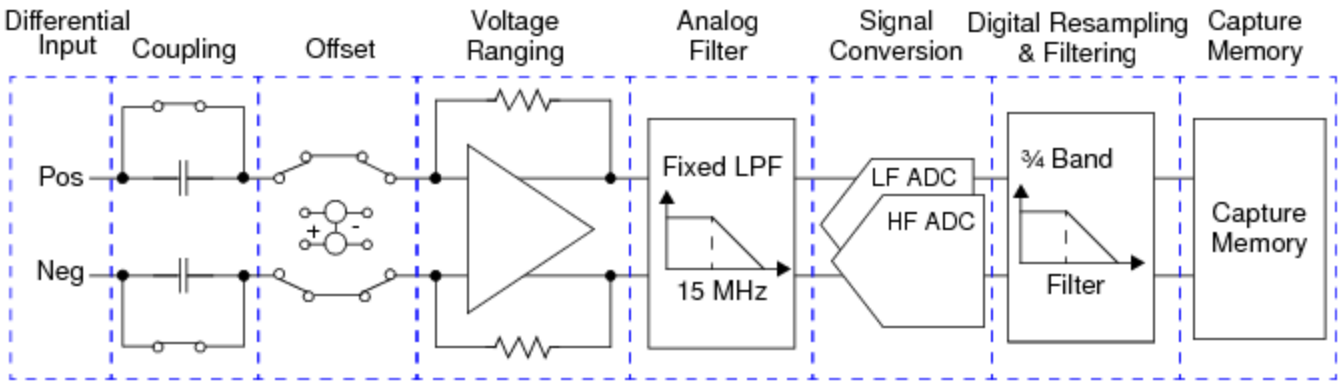
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BBAC Capture Theory of Operation : BBAC Capture Signal Path

BBAC Capture Signal Path

This figure shows that the BBAC Capture is a differential instrument that receives analog input signals on its two pins (Pos and Neg) from the DUT. The signal is passed through an optional coupling and offset stage before the signal’s amplitude is scaled in the voltage ranging stage. An analog anti-aliasing filter eliminates any high frequency signal components and ensures the BBAC Capture target bandwidth of 15MHz. The signal is then sampled at a fixed sample rate of 100MHz using analog-to-digital converters.



BBAC Capture Instrument Components

After the fixed sample rate conversion, the data passes through a digital resampling filter which resamples the capture data at the programmed sample rate. The data is then stored in the instrument’s capture memory until the instrument performs an instrument initiated move (IIM). The IIM moves the data to the support board’s memory or to the host computer for signal analysis. Details of this process are described in the following topics:

- Differential Input
- Coupling and Offset

- [Voltage Ranging](#)
- [Analog Low-Pass Filter](#)
- [Signal Conversion](#)
- [Digital Resampling and Filtering](#)
- [Capture Memory](#)

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BBAC Capture Theory of Operation : BBAC Capture Signal Path : Differential Input

Differential Input

An input signal enters the BBAC Capture instrument as an analog signal from the DUT. The signal is an arbitrary waveform that must have an amplitude less than the programmed voltage range and frequency components less than the instrument’s bandwidth to be accurately captured. Because the BBAC Capture is a differential instrument it is always associated with two physical wires.

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BBAC Capture Theory of Operation : BBAC Capture Signal Path : Coupling and Offset

Coupling and Offset

The coupling and offset circuitry of the BBAC Capture allows you to maximize the dynamic range of the instrument by removing DC offset from the capture signal. By removing the differential offset of the signal you can achieve the optimal dynamic range of the capture range selected. Additionally, you can remove common mode offset from the signal (if necessary) to ensure that the signal on each pin with respect to the BBAC Channel Card Common (CCC) does not exceed the instrument’s compliance.

For more information on the BBAC Capture compliance value, see the BBAC15 Specifications.

The captured signal can be DC or AC coupled. When DC coupled, the signal’s AC and DC components are passed. The advantage of DC coupling is that the instrument settling time is less because the high-pass filter used for AC coupling is bypassed. When AC coupling is applied, the DC component of the signal is blocked and only the AC components are passed. A precharging circuit is available when using AC coupling, which reduces the settling time associated with the AC coupling high-pass filter. The precharge circuit works to quickly match the DC voltage on either side of the capacitor to minimize settling time.

You can apply the DC offset removal circuitry to” the positive pin, the negative pin, or both. When the DC offset removal is applied to both pins, common mode offsets can be removed.

For more information, see:

- BBAC Capture Instrument Components
- Precharge

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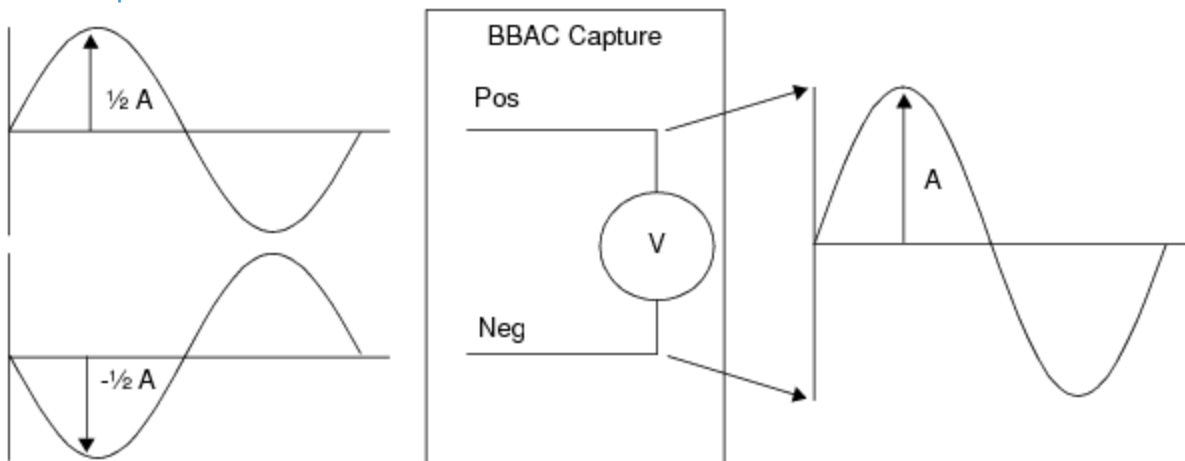
BBAC Capture Theory of Operation : BBAC Capture Signal Path : Voltage Ranging

Voltage Ranging

The range specifies the maximum or minimum voltage that can be applied between the positive and negative pins, or differential voltage. For example, if the voltage between the positive and negative pins is 1.28V, the range must be set to 1.28V or higher to accommodate the signal. Setting the capture to a range smaller than the maximum voltage causes the capture to clip the signal, resulting in an overrange alarm.

Another limitation on the capture is that the voltage on the positive or negative pins must be less than the instrument compliance ($\sim 7\text{V}$ with respect to the channel card common). For more information, see the BBAC15 Specifications.

The 7V constraint applies to the single-ended voltage between each pin and the channel card common, whereas the range applies to the differential voltage between the positive and negative. If the instrument is connected to a balanced differential signal source of 10.24Vpk amplitude (as shown in the following figure, $A = 10.24\text{V}$), the positive and negative pins will never exceed 5.12V with respect to channel card common, which is smaller than the 7V limit.

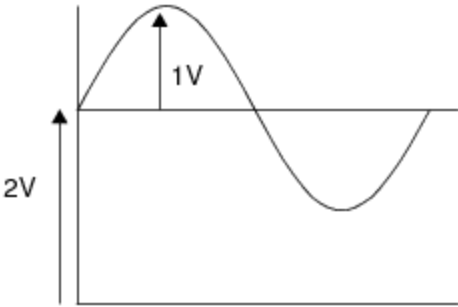


Differential Signal Capture

Valid BBAC Capture ranges are: 10.24V, 7.64V, 5.12V, 3.62V, 2.56V, 1.81V, 1.28V, 0.905V, 0.64V, 0.452V, 0.32V, 0.226V, 0.16V, and 0.113V.

In the figure, the input signal is a balanced differential sine wave with amplitude A. The maximum or minimum instantaneous voltage ever presented between the positive and negative pin is at the peaks, A volts. The capture range must therefore be set to a value greater than or equal to A. This is not the same as the peak-to-peak voltage of the sine wave, which is 2*A. Peak-to-peak refers to the maximum voltage difference encountered over time.

When dealing with ranging issues, the instrument is only concerned about the instantaneous voltage. The maximum Vpp of a captured sine wave is only twice the range voltage when the signal has no differential offset. For example, to capture the differential signal shown in the following figure, the instrument should be set to a range of at least 3.62V to avoid saturating the capture (assuming the baseline circuit is not used). The 1.28V and 2.56V ranges would be too small for this signal because the combination of the AC and DC components creates a signal that has a maximum instantaneous voltage differential of 3V.



Signal with 2V DC Offset

As a final example, consider a loopback between a BBAC Source and BBAC Capture, which sources and captures a signal of 2Vpp, using no offset and the best possible range. The 2Vpp signal corresponds to a 1V amplitude signal, so the source range must be set to the 1.28V range to optimally generate this signal. Since the maximum differential voltage generated by the source is 1V, the capture range must also be set to 1.28V.

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BBAC Capture Theory of Operation : BBAC Capture Signal Path : Analog Low-Pass Filter

Analog Low-Pass Filter

The scaled input signal passes through a fixed low-pass filter to eliminate any high frequency components. Attenuating frequency components beyond 15MHz minimizes the effects of aliasing from the sampling process of the instrument’s ADCs. This ensures that the instrument meets its performance specifications.

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BBAC Capture Theory of Operation : BBAC Capture Signal Path : Signal Conversion

Signal Conversion

The filtered signal is passed to the analog-to-digital converters (ADCs), which effectively sample the analog waveform at a frequency of 100Msps. The converted analog signals are separated into high (~3 to 15MHz) and low (~DC to 3MHz) frequency components and are passed on to the digital resampling and filter section for recombination.

Clock Resolution

The BBAC has a period-based clock. Resolution is defined in terms of period resolution and therefore is frequency dependent. The BBAC frequency is governed by the equation:

$$\text{BBAC Clock Frequency} = \frac{100 \text{ MHz} \times 2^{40}}{2^{40} \times (I + J)}$$

(1)

Where: $I \geq 1, 0 < J < 2^{40}$

The 40 bits of clock resolution allows both the BBAC Source and BBAC Capture instruments to achieve any desired frequency to a degree of precision that far exceeds the digital subsystem. While the BBAC is not always guaranteed to be coherent with the digital subsystem in the strict sense (a perfect integer ratio relationship), any source or measurement error introduced due to coherence error will be less than can be measured by the BBAC analog performance.

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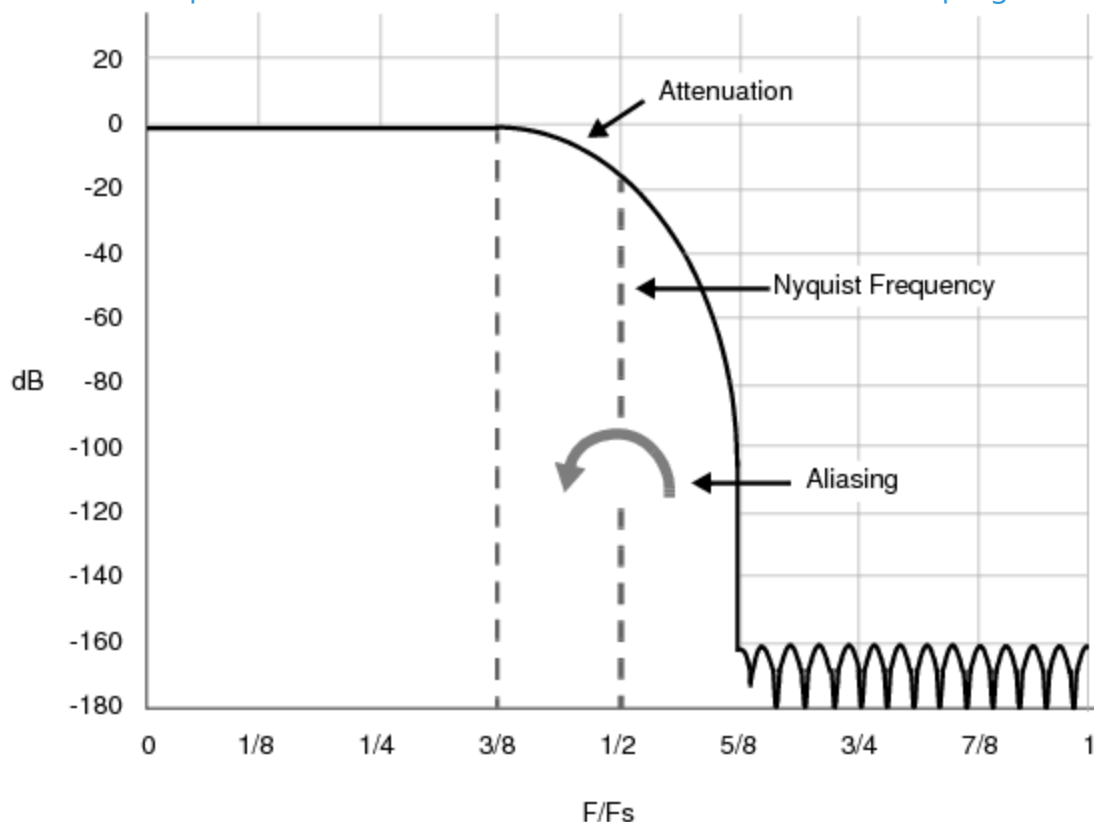
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BBAC Capture Theory of Operation : BBAC Capture Signal Path : Digital Resampling and Filtering

Digital Resampling and Filtering

The high and low frequency digital signals are recombined into one signal by a digital crossover filter. This fixed data rate signal is then resampled at the programmed sample rate through stages of interpolation and decimation. The signal is then passed through a 3/4 band filter, which removes any aliasing that was introduced by the resampling process.

The impact of the 3/4 band filter on the capture signal is important. The following figure shows that the frequency response shape of the 3/4 band filter is always the same, while the bandwidth varies with the sample rate. The bandwidth is $0.75 * F_s/2$, where F_s is the programmed sampling rate.



3/4 Band Filter

The spectrum shown above is between $3/8 \cdot F_s$ and $F_s/2$ should not be included in any calculations because:

- The filter has started to roll off, so the frequency response is not flat.
- Any signals or noise present in the range of $F_s/2$ to $5/8 \cdot F_s$ are only partially attenuated and will alias back into this part of the spectrum.

To avoid aliasing, choose F_s using the following procedure:

1. Let BW = highest frequency of interest for the test.
2. Choose an F_s such that $F_s \geq 8/3 \cdot BW$.
3. Restrict all spectral analysis (for example, SNR) to the region of DC to BW .

Note that Mixed Signal Workshop automatically enforces step 2. You need only provide the required bandwidth to the Solver.

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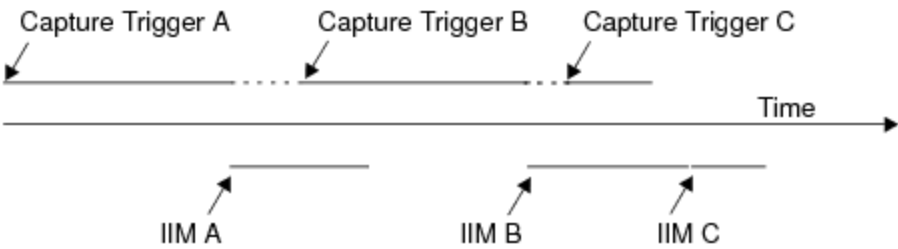
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BBAC Capture Theory of Operation : BBAC Capture Signal Path : Capture Memory

Capture Memory

Digital values from the 3/4 band output filter are routed to a 1,048,064-sample capture memory. The BBAC Capture uses an instrument initiated move (IIM) to automatically move the captured data from the instrument's capture memory (CMEM) to the memory in either a system DSP computer or the host computer for signal analysis. Each capture begins where the previous capture ended. As soon as a capture completes, the next capture can begin.

This figure shows three consecutive captures. As soon as the first capture completes it is automatically moved through IIM for processing, and the second capture begins while the first is moving. The third capture then begins and completes while the second is still moving. As soon as a capture finishes moving, its space in CMEM is freed.



Capture Sequence and Instrument Initiated Move (IIM)

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BBAC Capture Theory of Operation : Connectivity Options

Connectivity Options

The following topics describe the options for connecting a BBAC Capture instrument to various resources through the DIB:

- | | |
|---|---|
| • | Differential Device Output Connections |
| • | Single-Ended Device Output Connections |
| • | Shared Resources |
| • | DIB Access |
| • | Multiple Resources |
| • | Using RefGround to Connect a DUT Pin to Instrument Ground |

☒ **Note:**

DGS pins are intended for use as a ground reference for the PPMUs and not for BBAC Capture instruments. Misapplication of a DGS connection can result in the instrument not meeting its performance specifications.

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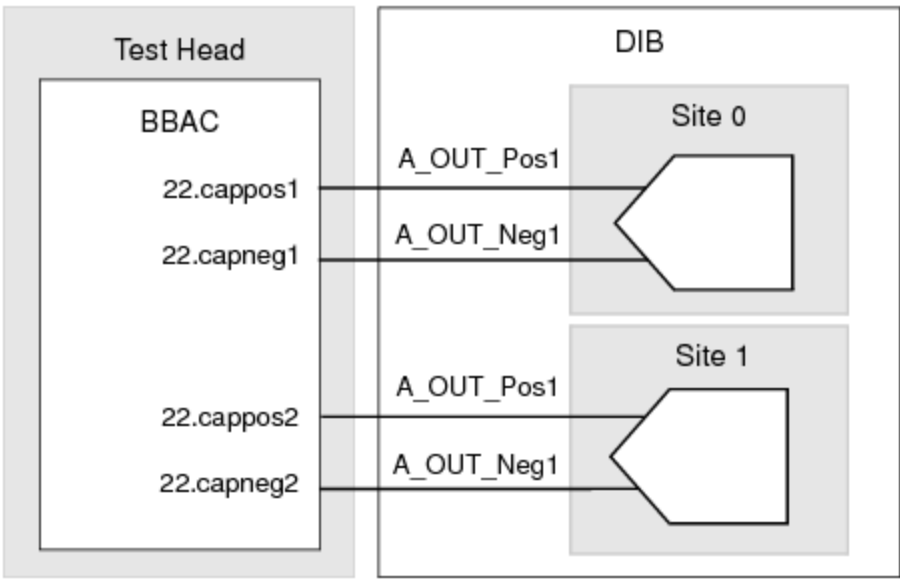
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BBAC Capture Theory of Operation : Connectivity Options : Differential Device Output Connections

Differential Device Output Connections

The figure shows a differential connection between a BBAC Capture instrument and a device with differential outputs.



BBAC Capture Connected Differentially to the DUT

Pin Map for BBAC Capture Differential Connections

Group Name	Pin Name	Type
	A_OUT_Pos1	Analog
	A_OUT_Neg1	Analog

Channel Map for BBAC Capture Differential Connections

Pin Name	Package Pin	Type	Site 0	Site 1
A_OUT_Pos1		BBACCapPos	22.cappos1	22.cappos2
A_OUT_Neg1		BBACCapNeg	22.capneg1	22.capneg2

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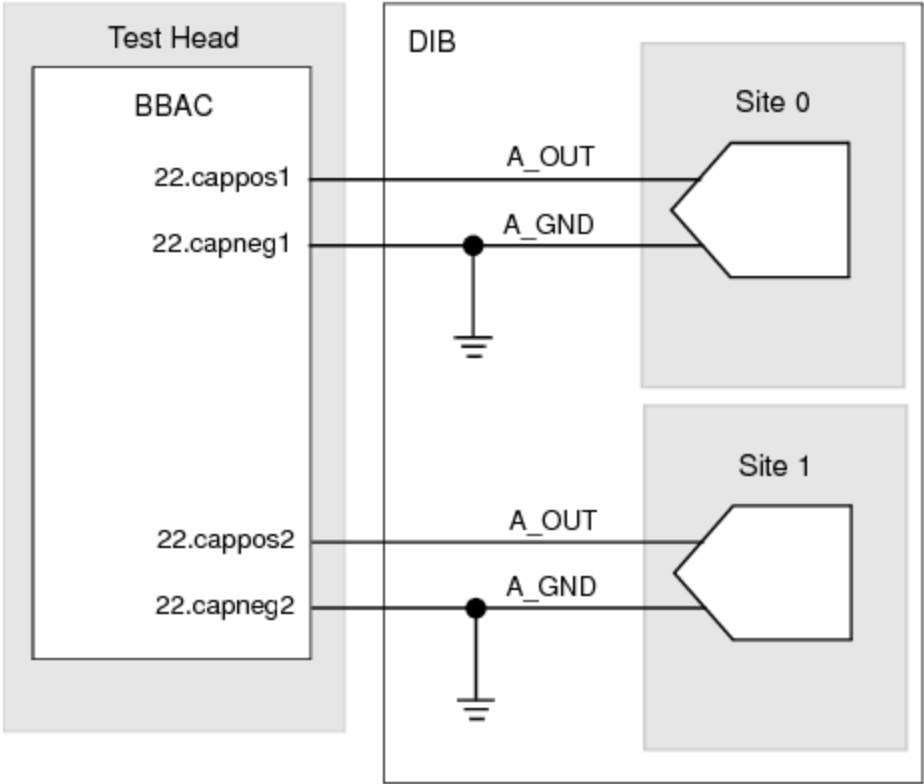
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BBAC Capture Theory of Operation : Connectivity Options : Single-Ended Device Output Connections

Single-Ended Device Output Connections

For single-ended device outputs, DIB circuitry is not required for single-ended to differential conversion, as shown in the following figure.



Single-Ended Device Output Configuration

Pin Map for BBAC Capture Single-Ended Connection

Group Name	Pin Name	Type
------------	----------	------

A_OUT	Analog
A_GND	Gnd

Channel Map for BBAC Capture Single-Ended Connection

Pin Name	Package Pin	Type	Site 0	Site 1
A_OUT		BBACCapPos	22.cappos1	22.cappos2
A_GND		Gnd		
A_GND:M		BBACCapNeg	22.capneg1	22.capneg2

The most common DUT/instrument connectivity for single-ended devices is to connect the BBAC Capture Positive pin to the device output, and the BBAC Capture Negative to the DUT/DIB ground. Note that the positive and negative pins could be swapped. For consistency, however, connect the positive pin to the device output. See the pin map above.

Note in the channel map above that the BBAC Capture Negative pin is shown as connected to the device ground pin as a multiple resource. This is done not for programming the BBAC Capture (which can be programmed using only the positive pin), but for control of the PPMU behind the negative pin. If the PPMU is not required, then you need not list the BBAC Capture Negative pin in the Channel Map.

If a device provides a more suitable reference for the device output than ground, you can connect the capture negative pin to this pin. The primary advantage of connecting to a device reference is that it can provide a smaller DC offset between the positive and negative pins. This allows for optimal use of the dynamic range without using the DC baseline removal circuitry which adds noise to the capture measurement.

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BBAC Capture Theory of Operation : Connectivity Options : Shared Resources

Shared Resources

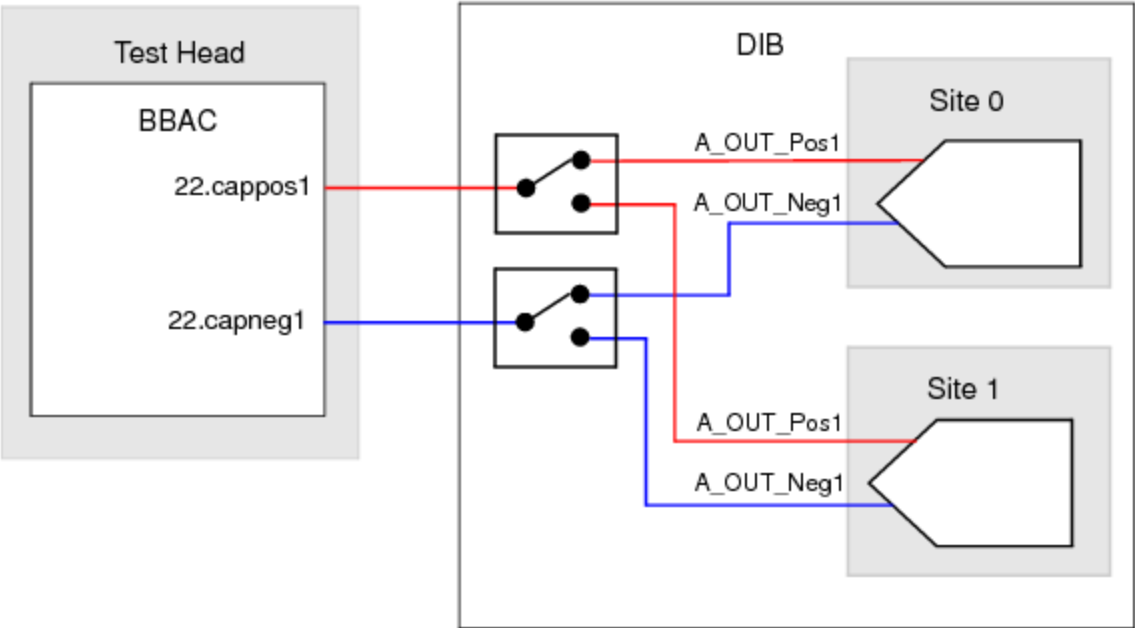
UltraFLEX allows a single instrument to be shared by more than one DUT pin or more than one site in a multisite test environment. However, sharing resources requires that the same instrument channel or pin be used more than once in the channel map. To avoid conflicts, you must modify the standard channel map entries that indicate your intentions.

The following sections describe sharing resources across the pins of the same device and across devices in different sites:

- Resources Shared Across Multiple Sites
- Shared Resources Within a Site

Resources Shared Across Multiple Sites

The figure shows that two or more sites can share a resource by indicating that the same tester channel is connected to more than one site. The first of the sharing sites is the shared reference site. Its tester channel is indicated in the standard way. All subsequent sharing sites use the keyword `site<n>` (for example, `site0`) to indicate the shared reference site with which the resource is being shared.



Resources Shared Across Multiple Sites

Pin Map for BBAC Capture Multiple Site Connections

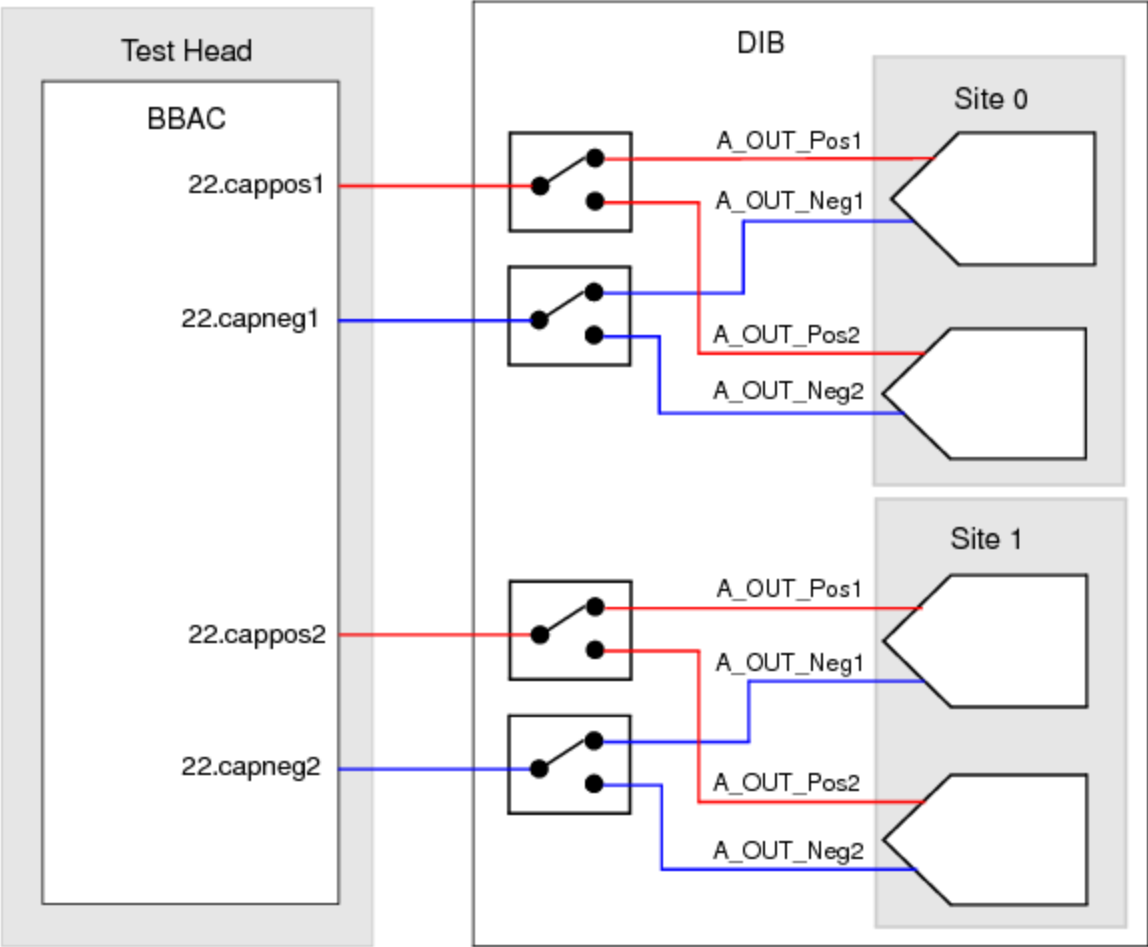
Group Name	Pin Name	Type
	A_OUT_Pos1	Analog
	A_OUT_Neg1	Analog

Channel Map for BBAC Capture Multiple Site Connections

Pin Name	Package Pin	Type	Site 0	Site 1
A_OUT_Pos1	BBACCapPos		22.cappos1	site0
A_OUT_Neg1	BBACCapNeg		22.capneg1	site0

Shared Resources Within a Site

The following figure shows that a pin can share a resource with other pins by indicating that the same tester channel is connected to both pins. The first pin encountered in the channel map that shares a tester resource designates the tester channel in the standard way. All subsequent pins that share this resource then use the following syntax format to specify their tester channel:
S:<shared-reference-pin>



Shared Resources Within a Site

Pin Map for BBAC Capture Shared Resource Connections

Group Name	Pin Name	Type
	A_OUT_Pos1	Analog
	A_OUT_Neg1	Analog
	A_OUT_Pos2	Analog
	A_OUT_Neg2	Analog

Channel Map for BBAC Capture Shared Resource Connections

Pin Name	Package Pin	Type	Site 0	Site 1
A_OUT_Pos1	BBACCapPos		22.cappos1	22.cappos2
A_OUT_Neg1	BBACCapNeg		22.capneg1	22.capneg2
A_OUT_Pos2	BBACCapPos		s:A_OUT_Pos1	

A_OUT_Neg2

BBACCapNeg

s:A_OUT_Neg1

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BBAC Capture Theory of Operation : Connectivity Options : DIB Access

DIB Access

A test often requires several instrument types to be connected to a DUT pin in a test program. IG-XL supports this requirement in two ways:

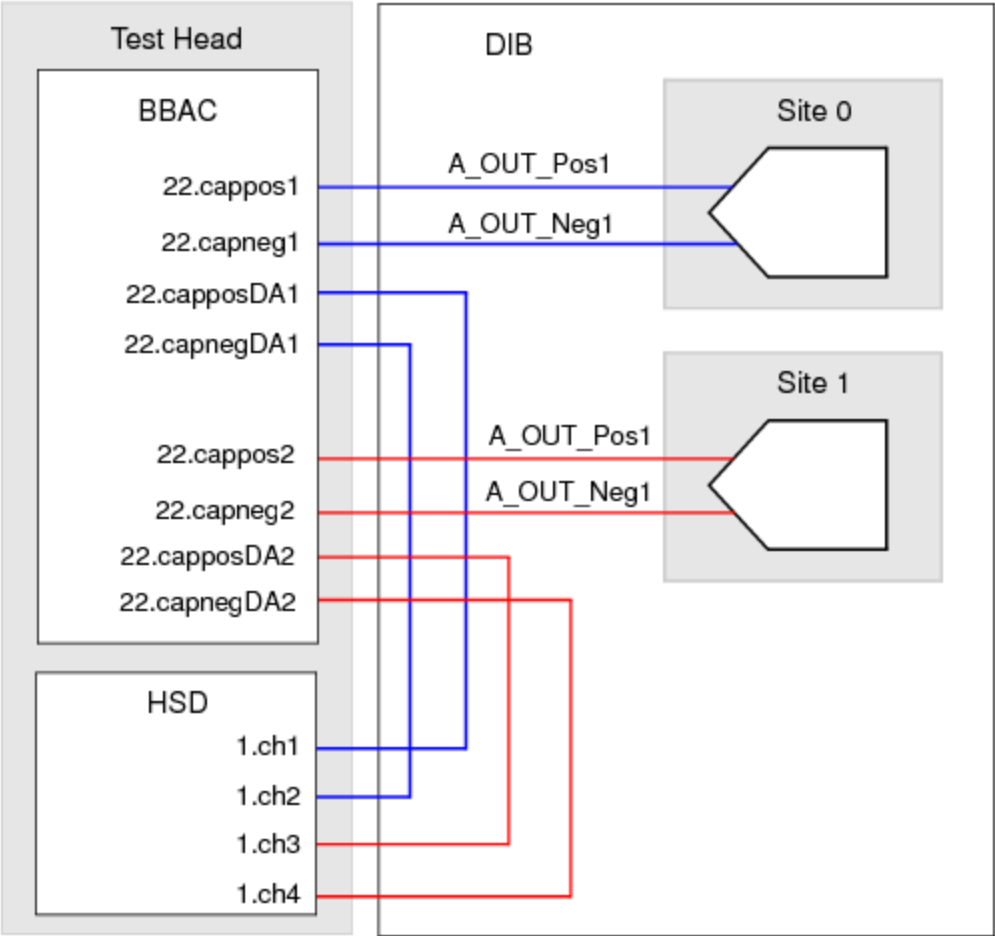
- Directly connecting multiple resources to the DUT pin. See Multiple Resources.
- Using the DIB access feature to connect the DUT pin to one instrument board through another instrument board

When a device pin needs multiple or different resources, you can use DIB access to minimize the number of DIB relays that are required and allow for a simpler programming model. Once the channel map is set up to indicate the DIB connectivity, IG-XL addresses the correct resource based solely on the pin name. You need not create a virtual pin to reference the different resources.

- BBAC Capture as Primary Resource
- BBAC Capture as Secondary Resource

BBAC Capture as Primary Resource

When a BBAC Capture is used as the primary resource for the device pin, its corresponding DIB access pins are connected (using DIB traces) to another instrument’s resources. The figure shows that HSD resources are delivered as secondary resources to the device pin using DIB access.



DIB Access Connection, BBAC Capture as Primary Resource

Pin Map for BBAC Capture DIB Access as Primary Resource

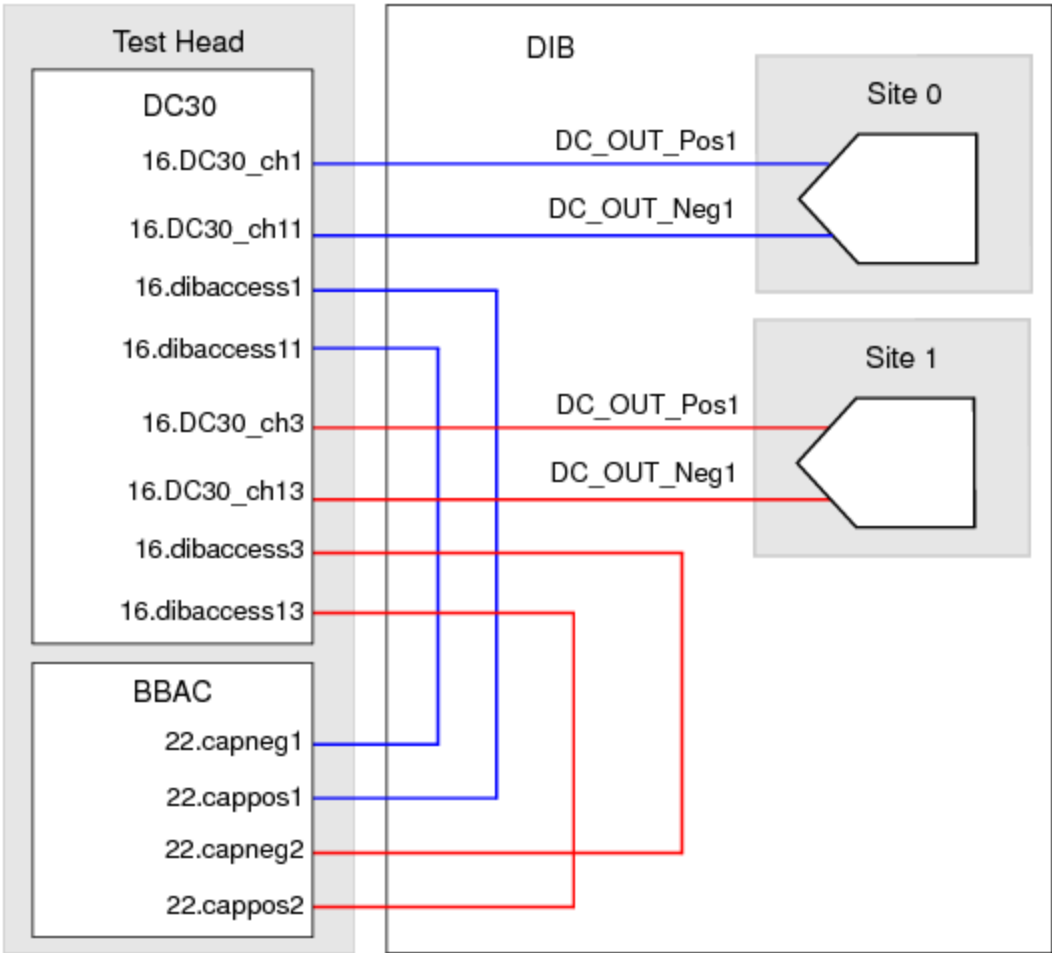
Group Name	Pin Name	Type
	A_OUT_Pos1	Analog
	A_OUT_Neg1	Analog

Channel Map for BBAC Capture DIB Access as Primary Resource

Pin Name	Package Pin	Type	Site 0	Site 1
A_OUT_Pos1		BBACCapPos	22.cappos1	22.cappos2
A_OUT_Pos1:DA		I/O	1.ch1	1.ch3
A_OUT_Neg1		BBACCapNeg	22.capneg1	22.capneg2
A_OUT_Neg1:DA		I/O	1.ch2	1.ch4

BBAC Capture as Secondary Resource

The DIB access use model for the BBAC Capture as a secondary resource for a device pin is identical to when it is a primary resource. The only differences are the DIB connectivity and the channel map setup. The figure shows a BBAC Capture is delivered as a secondary resource to the device pins using DIB access.



BBAC DIB Access, BBAC Capture as a Secondary Resource

Note:

This figure may not accurately reflect the DIB access capability of the DC30.

Pin Map for BBAC Capture DIB Access as Secondary Resource

Group Name	Pin Name	Type
	DC_OUT_Pos	Analog
	DC_OUT_Neg	Analog

Channel Map for BBAC Capture DIB Access as Secondary Resource

Pin Name	Package Pin	Type	Site 0	Site 1
----------	-------------	------	--------	--------

DC_OUT_Pos1	DCVI	16.sense1	16.sense3
DC_OUT_Pos1:DA	BBACCapPos	22.cappos1	22.cappos2
DC_OUT_Neg1	DCVI	16.sense11	16.sense13
DC_OUT_Neg1:DA	BBACCapNeg	22.capneg1	22.capneg2

 Note:

The DIB access row in the channel map must immediately follow the pin that it represents.





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BBAC Capture Theory of Operation : Connectivity Options : Multiple Resources

Multiple Resources

A test often requires several instrument types to be connected to a DUT pin in a test program. IG-XL supports this requirement in two ways:

- [Directly connecting multiple resources to the DUT pin](#)
- [Using the DIB access feature to connect the DUT pin to one instrument board through another instrument board. See DIB Access.](#)

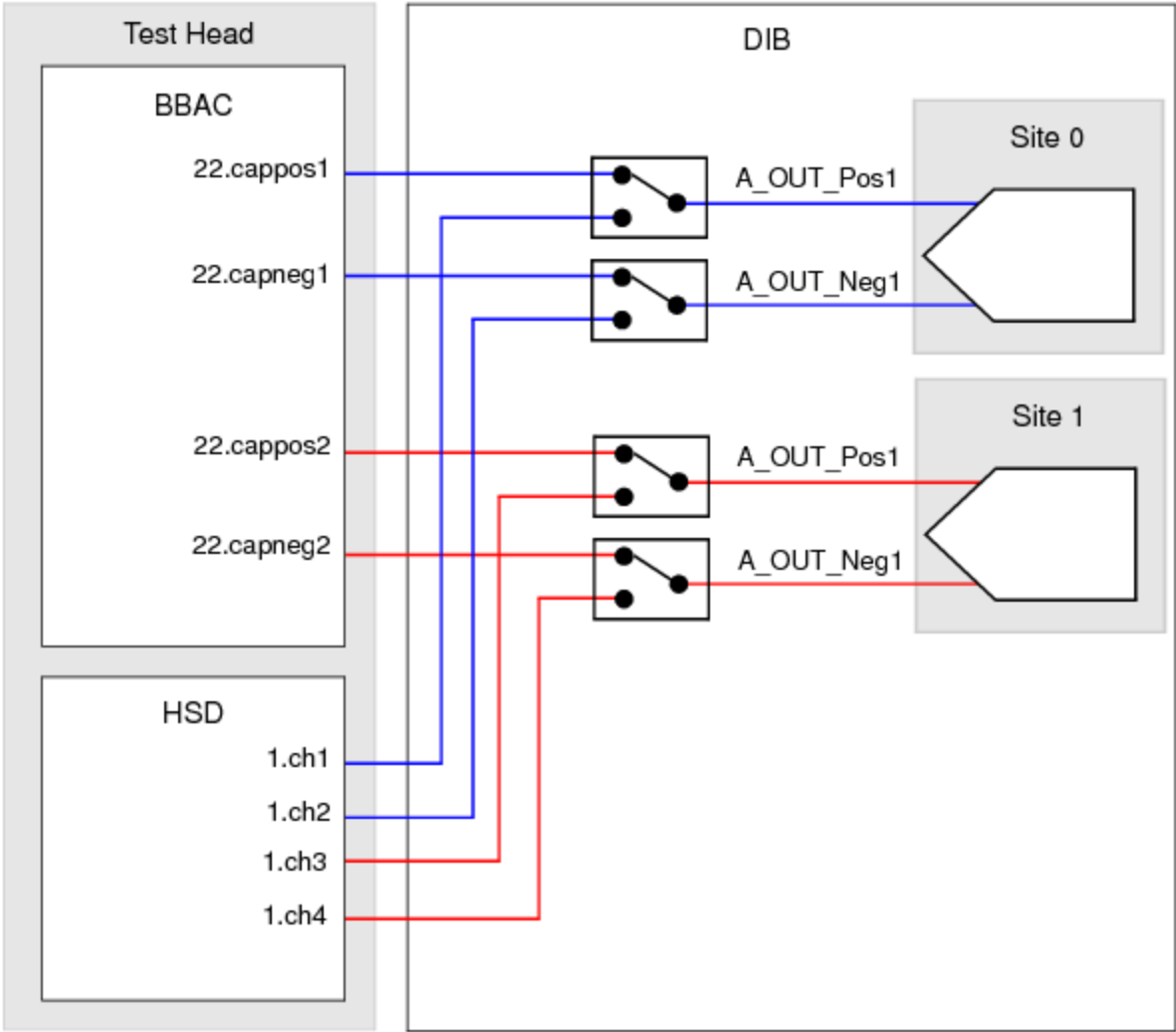
Multiple resources are similar to DIB access with the primary difference being the implied DIB connectivity. Like DIB access, once the channel map is set up to indicate how instrument resources have been assigned, IG-XL automatically addresses the correct resource based solely on the pin name. You need not create a “virtual” pin to reference the different resources. In most cases of multiple resources, you provide DIB relays to manage instrument connectivity.

[Note from BBAC Capture as a Primary Resource](#) and [BBAC Capture as Secondary Resource](#) that the DIB connectivity is often the same for the BBAC Capture as a primary and secondary resource. The significance of the ordering of the resources in the channel map is when both resources have similar instrumentation (and more importantly language). For example, in the case shown in [BBAC Capture as a Primary Resource](#), while both the BBAC pins and the HSD pins have PPMU resources available, because the BBAC is listed as the primary resource, its PPMUs are used.

- [BBAC Capture as Primary Resource](#)
- [BBAC Capture as Secondary Resource](#)

BBAC Capture as Primary Resource

The figure shows an example of a device pin using multiple resources. In this example, the BBAC Capture is used as the primary resource, and the HSD resources are delivered as secondary resources to the device pin using DIB relays. The channel map identifies which instrument is the primary instrument. See [Multiple Resources and DIB Access](#) in the DataTool section for more information about setting a Channel Map sheet for multiple resource programming.



BBAC Capture as a Primary Resource

Pin Map for BBAC Capture as Primary Resource

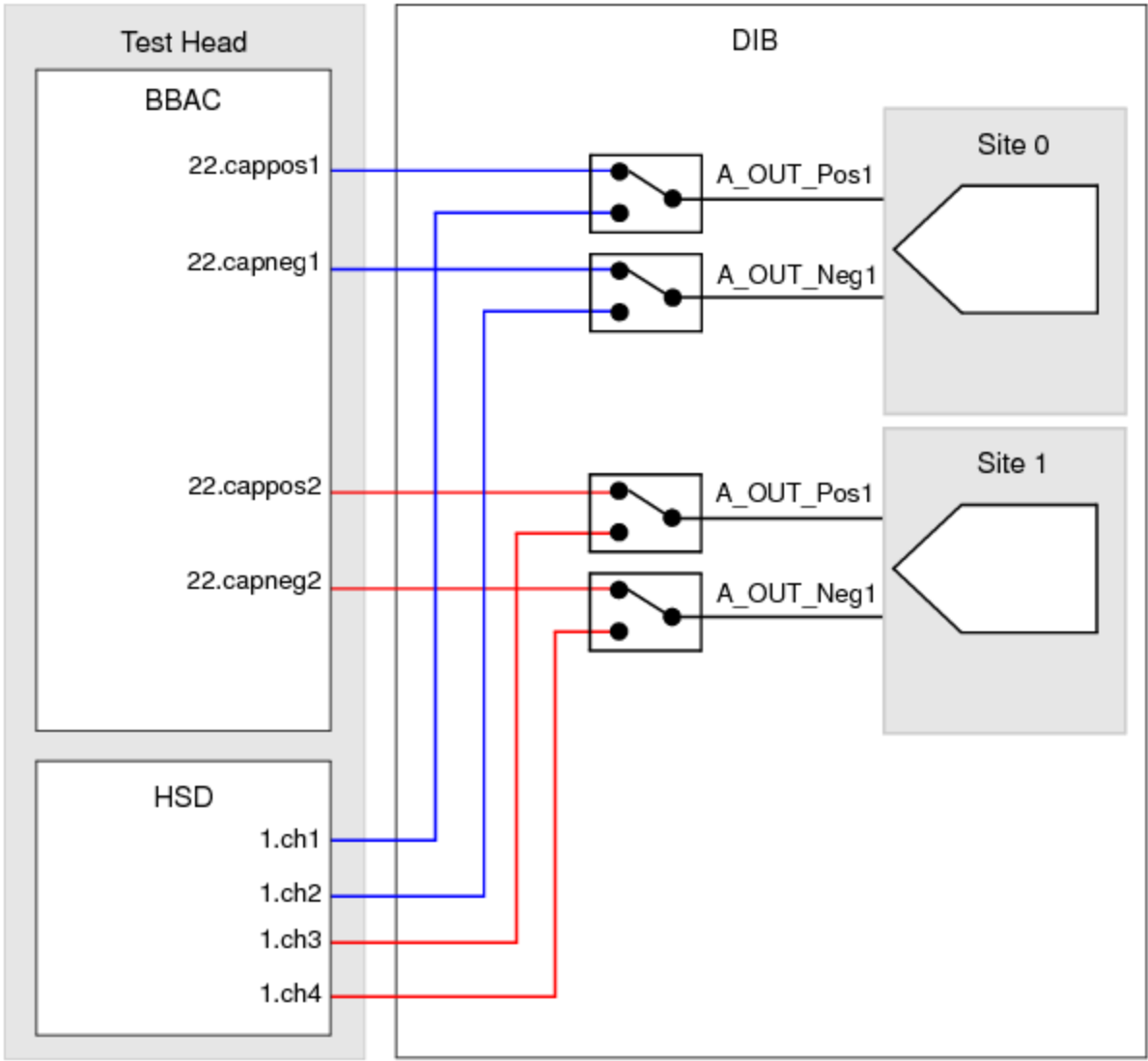
Group Name	Pin Name	Type
	A_OUT_Pos_1	Analog
	A_OUT_Neg_1	Analog

Channel Map for BBAC Capture as Primary Resource

Pin Name	Package Pin	Type	Site 0	Site 1
A_OUT_Pos1		BBACCapPos	22.cappos1	22.cappos2
A_OUT_Pos1:M		I/O	1.ch1	1.ch3
A_OUT_Neg1		BBACCapNeg	22.capneg1	22.capneg2
A_OUT_Neg1:M		I/O	1.ch2	1.ch4

BBAC Capture as Secondary Resource

The figure shows an example of a device pin using multiple resources. In this example, the BBAC Capture is used as the secondary resource, and the HSD resources are delivered as primary resources to the device pin using DIB relays. The channel map identifies which instrument is the primary instrument. See [Multiple Resources and DIB Access](#) in the DataTool section for more information about setting a Channel Map sheet for multiple resource programming.



BBAC Capture as Secondary Resource

Pin Map for BBAC Capture as Secondary Resource

Group Name	Pin Name	Type
	A_OUT_Pos1	I/O
	A_OUT_Neg1	I/O

Channel Map for BBAC Capture as Secondary Resource

Pin Name	Package Pin	Type	Site 0	Site 1
A_OUT_Pos1		I/O	1.ch1	1.ch3
A_OUT_Pos1:M		BBACCapPos	22.cappos1	22.cappos2
A_OUT_Neg1		I/O	1.ch2	1.ch4
A_OUT_Neg1:M		BBACCapNeg	22.capneg1	22.capneg2

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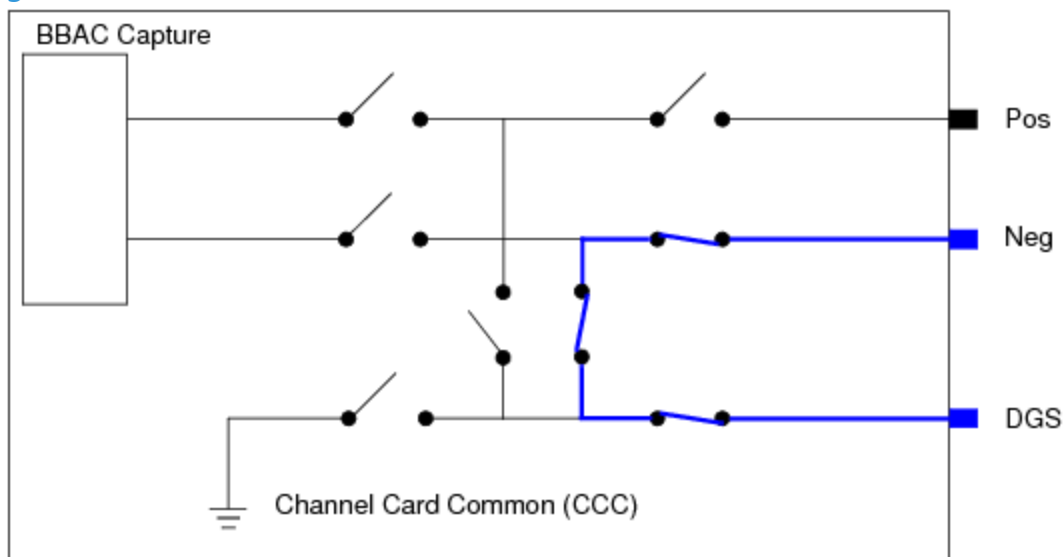
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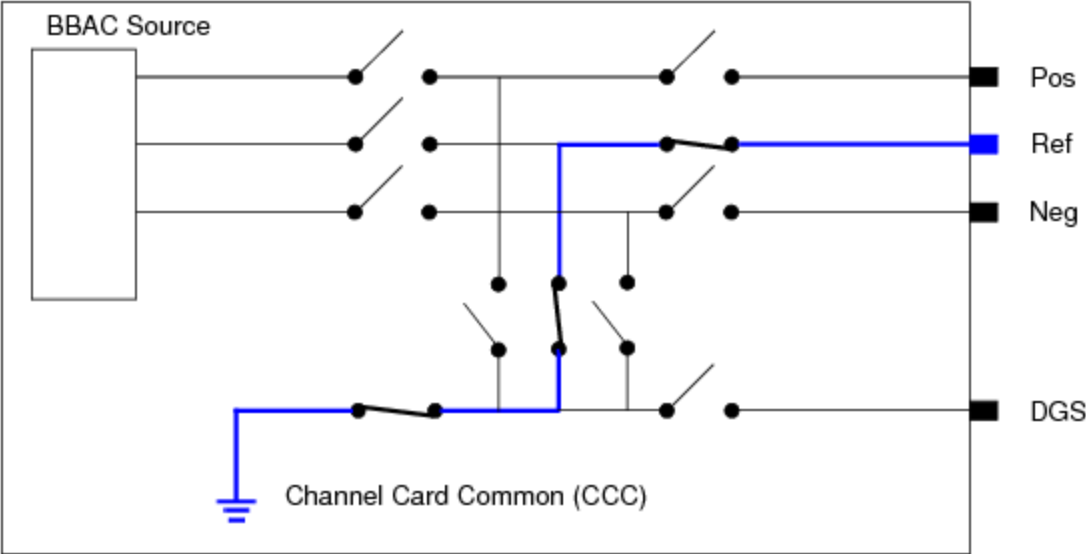
[BBAC Capture Theory of Operation : Connectivity Options](#) : Using RefGround to Connect a DUT Pin to Instrument Ground

[Using RefGround to Connect a DUT Pin to Instrument Ground](#)

RefGround is a logical instrument that allows you to ground DUT pins by connecting them to a BBAC Source or Capture's Channel Common Card Common (CCC) or Device Ground Sense (DGS). The following figures show how the RefGround instrument uses the BBAC hardware to connect to ground resources.



[RefGround Connects the DUT Pin to the BBAC Capture's DGS Resource](#)



RefGround Connects the DUT Pin to the BBAC Source CC Common Resource

The RefGround instrument has its own VBT syntax to connect or disconnect the DUT pins to or from the BBAC hardware. Use the following syntax to connect and disconnect to/from the instrument’s reference ground resources:

TheHdw.RefGround.Pins().Connect(tlCCGroundType)

TheHdw.RefGround.Pins().Disconnect(tlCCGroundType)

tlCCGroundType can be one of the following optional arguments:

tlCCGroundTypeDefault	The instrument’s default ground resource which is Channel Card Common.
tlCCGroundTypeCCC	The BBAC’s Channel Card Common ground resource.
tlCCGroundTypeDGS	The BBAC’s Device Ground Sense resource.

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BBAC Capture Theory of Operation : User View of DIB Pad Patterns

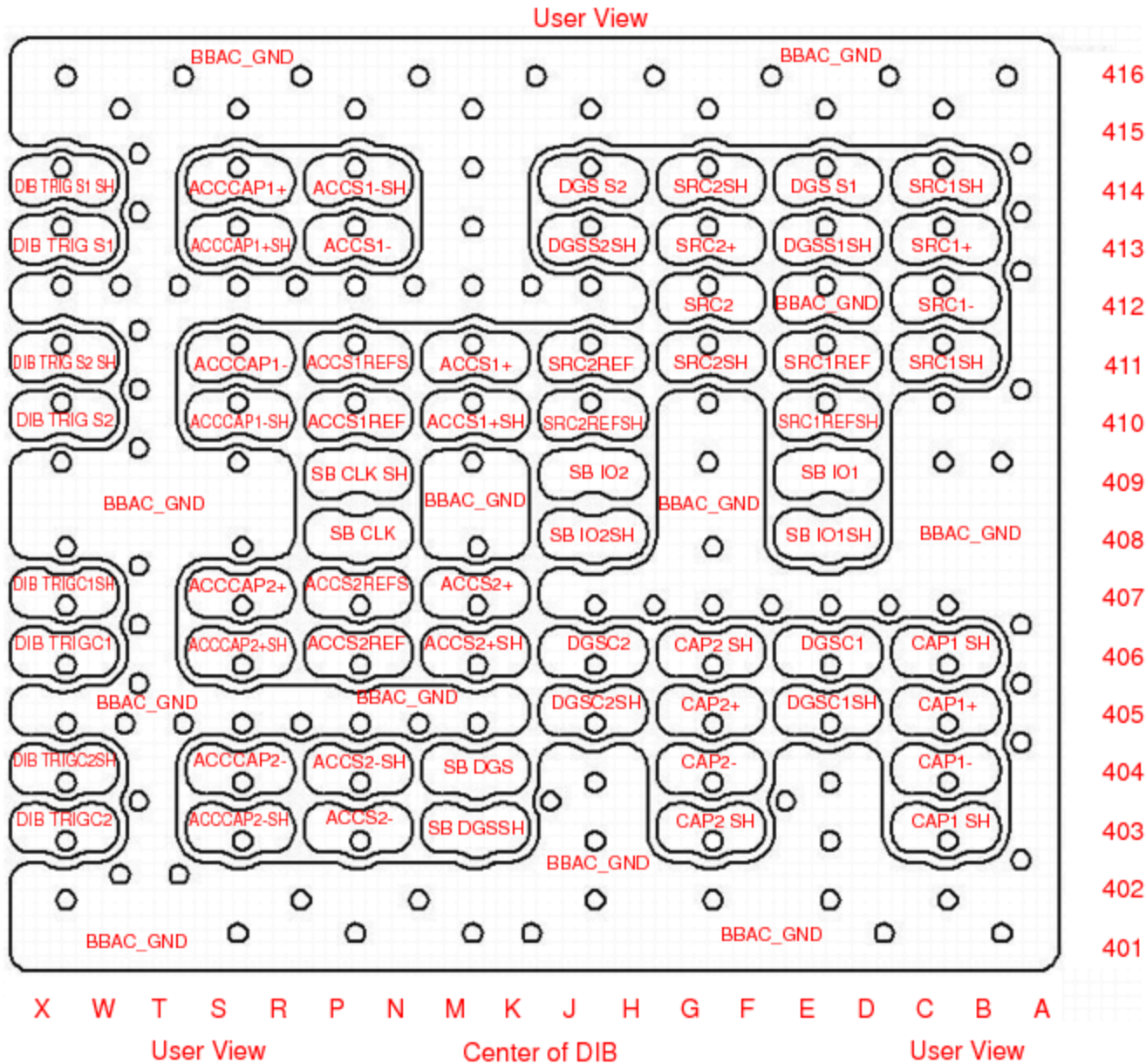
User View of DIB Pad Patterns

This section shows the BBAC pad patterns as seen from the user perspective, looking at the topside (device) of the DIB/HIB/PIB and towards the test system.

☑ Note:

When designing a DIB, use the tester view of the pad patterns (looking at the stiffener side of the DIB/HIB/PIB), which is given in [BBAC Signal Pad Pattern \(Tester View\)](#) because all the artwork and design kits use that view.

The following figure shows the BBAC pad pattern located on sidewalk 4.



BBAC Pad Pattern for Sidewalk 4 (User View)